General Chemistry I

Chemistry: The central science

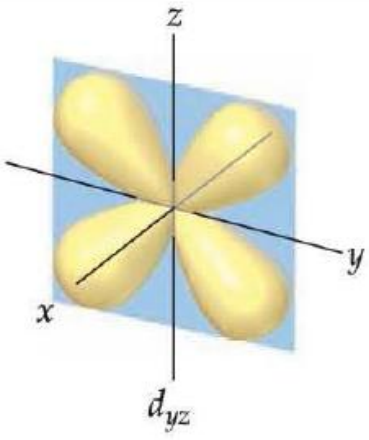
Phong D. Tran

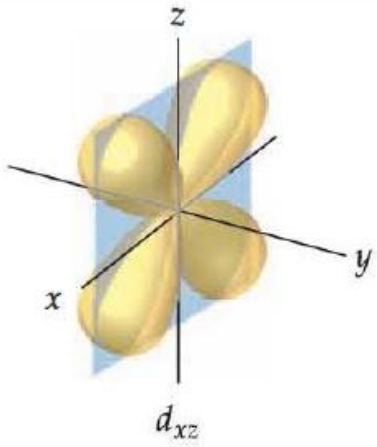
Department of Fundamental and Applied Sciences

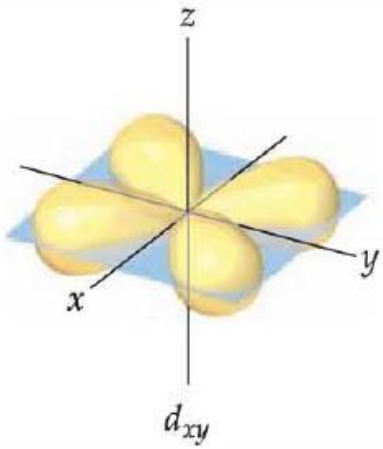
University of Science and Technology of Hanoi

Email: tran-dinh.phong@usth.edu.vn

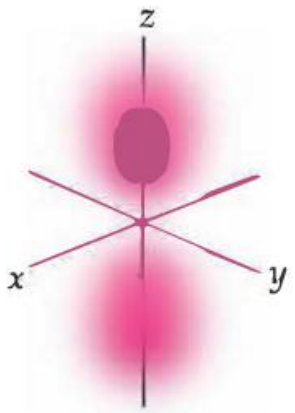
Lectures 5,6: Electronic structure of atoms



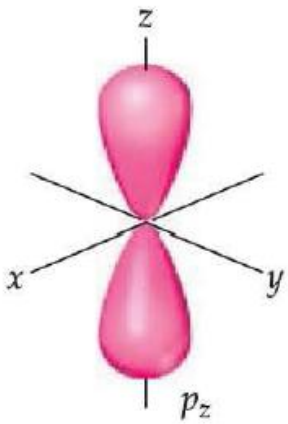


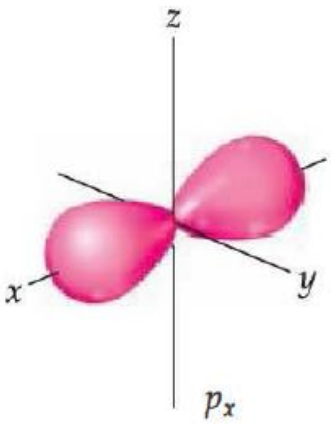


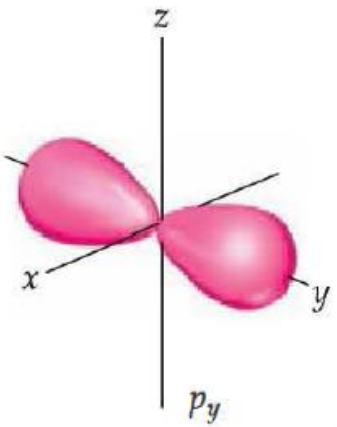
ELECTRONIC STRUCTURE OF ATOMS



(a)





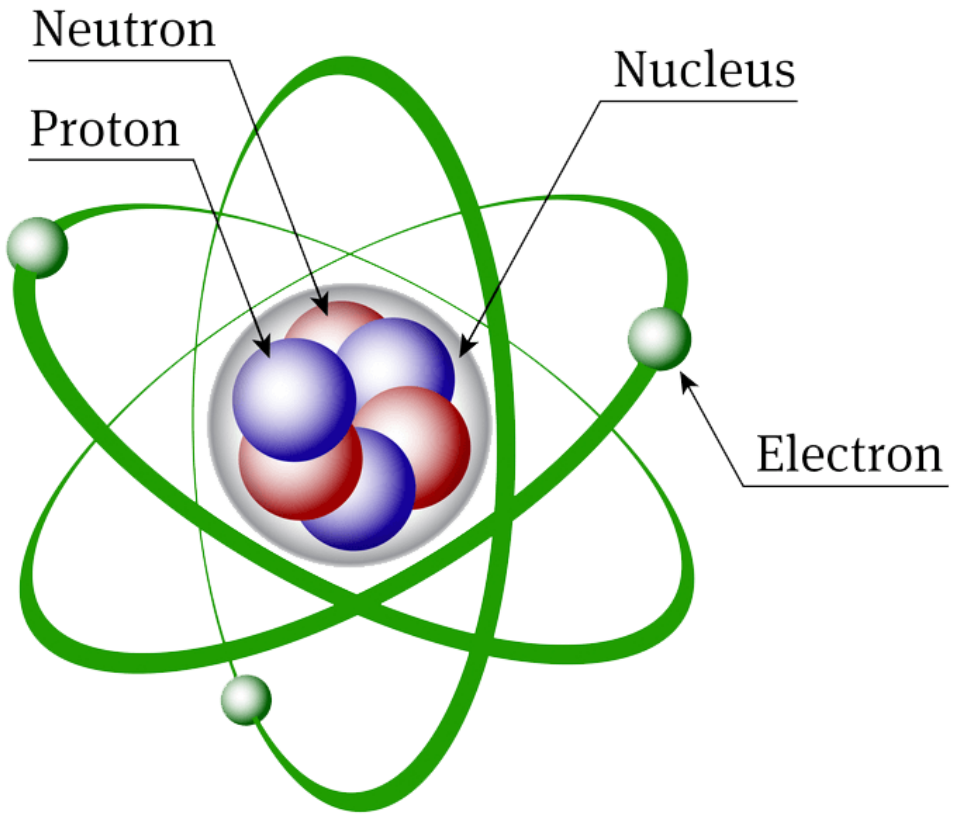


(b)

Contents

* Concepts of orbit and **orbital**
* Quantum numbers
* Electron configuration
* Electron configuration, energy and other physical properties

Atom is composed of electron, neutron and proton



How do electrons arrange around the nuclei?

Would all electrons bind with nuclei in the same manner?

Would all electrons have the same potential energy?

Would all electrons involve in chemical reactions?

Chemical properties are decided by electrons (valence electrons)

BUT not the nuclei (and core electrons)

ELECTRONIC STRUCTURE OF ATOMS

- Arrangement of electrons in space around the nucleus
- Electron energy

(*thus related to its physical and chemical activity*)

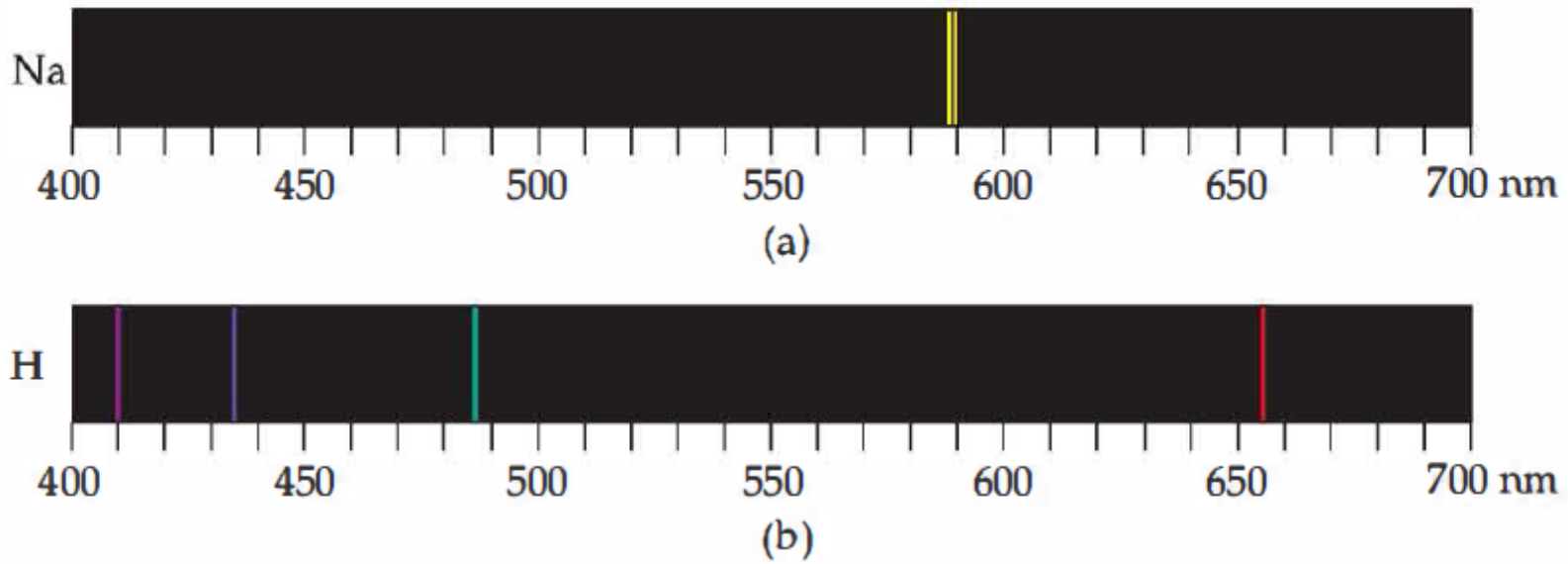
Given facts (*Observations*)

- Emission spectrum of atoms
- Periodic chemical properties of elements

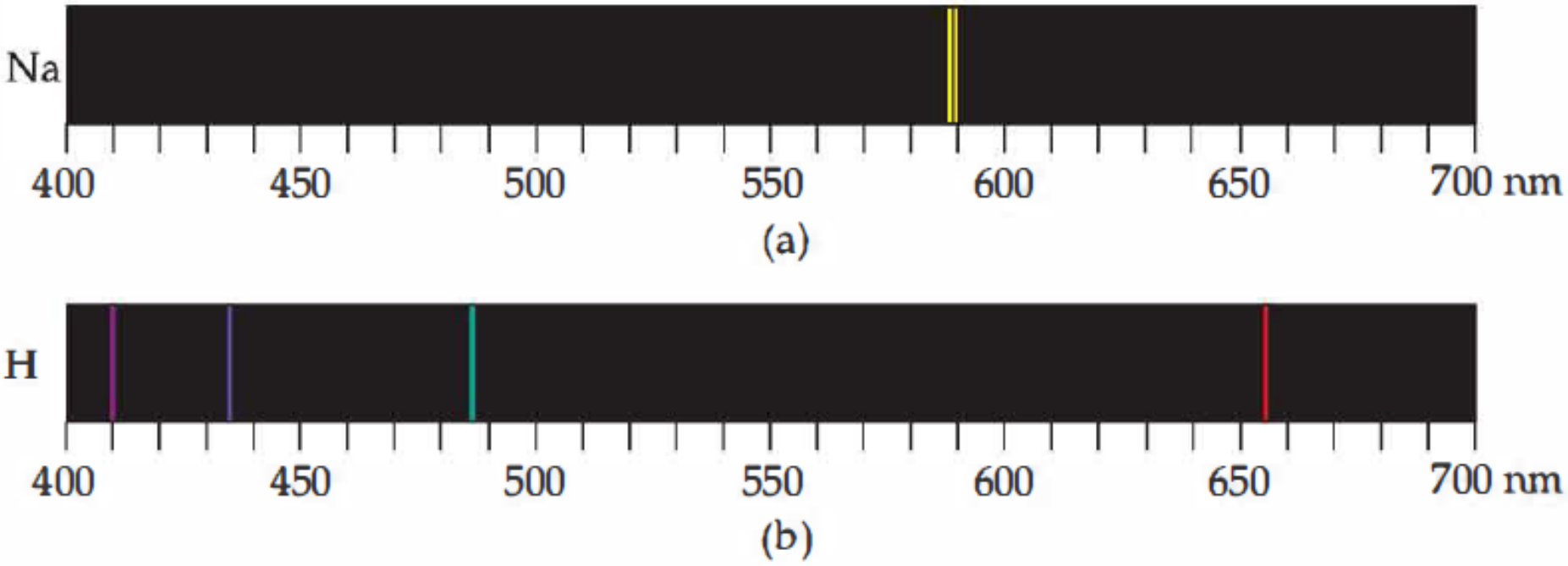
Continous spectrum: a rainbow of colors, containing light of all wavelengths



Line spectrum: containing only radiation of specific wavelengths



Atoms emit Line spectra BUT not continuous spectra



… Likely, electron could not stand at all energy level

….Classical physics may be not applicable

Different elements show (very) different chemical properties

He, Ne, Ar are inert gases

Li, Na, K are soft, very active (*violently reaction with water or O_2*)

Na



Na is kept in Kerosene oil



Electron(s) in different atoms would behave very differently.

They are not identical.

He balloons



Xmas in Chemistry Lab

(Ar balloons provide inert atmosphere for experiments)



ELECTRONIC STRUCTURE OF ATOMS

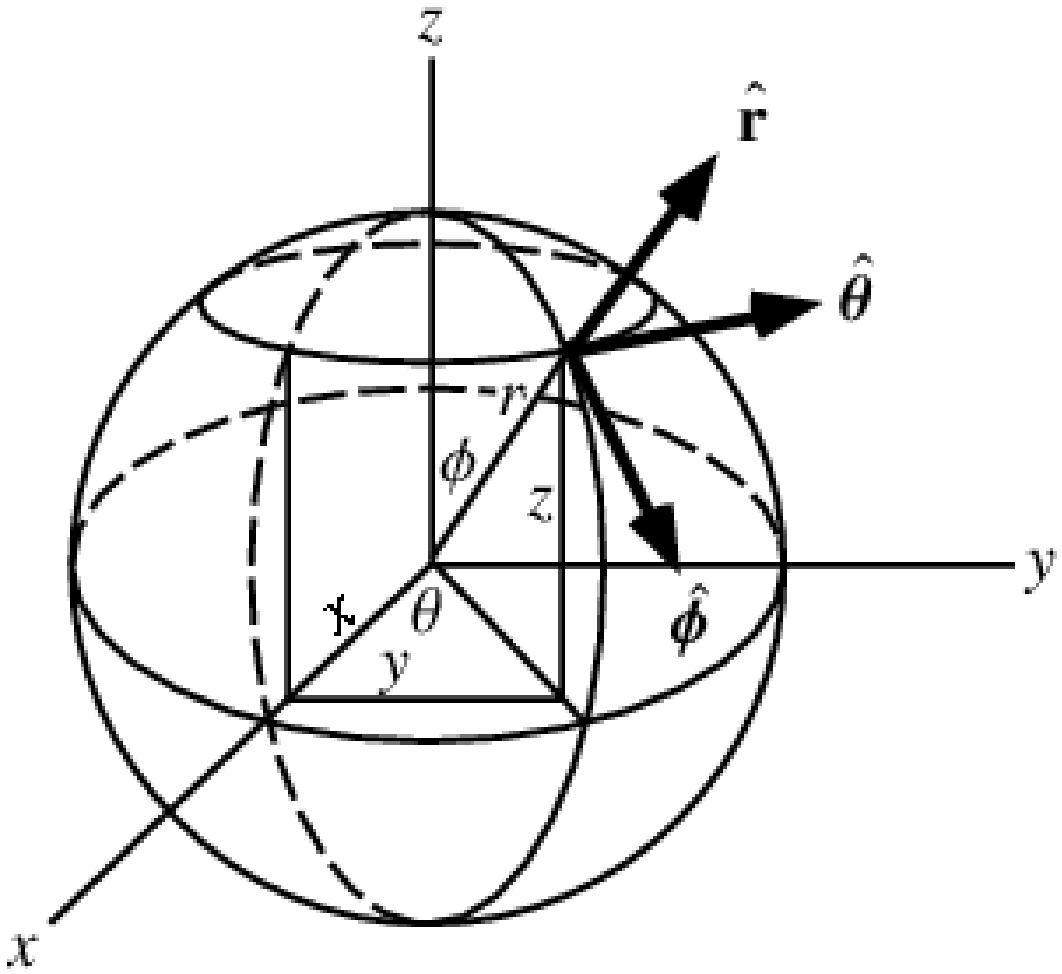
- Arrangement of electrons in space around the nucleus
- Electron energy

(*thus related to its physical and chemical activity*)

Given facts (*Observations*)

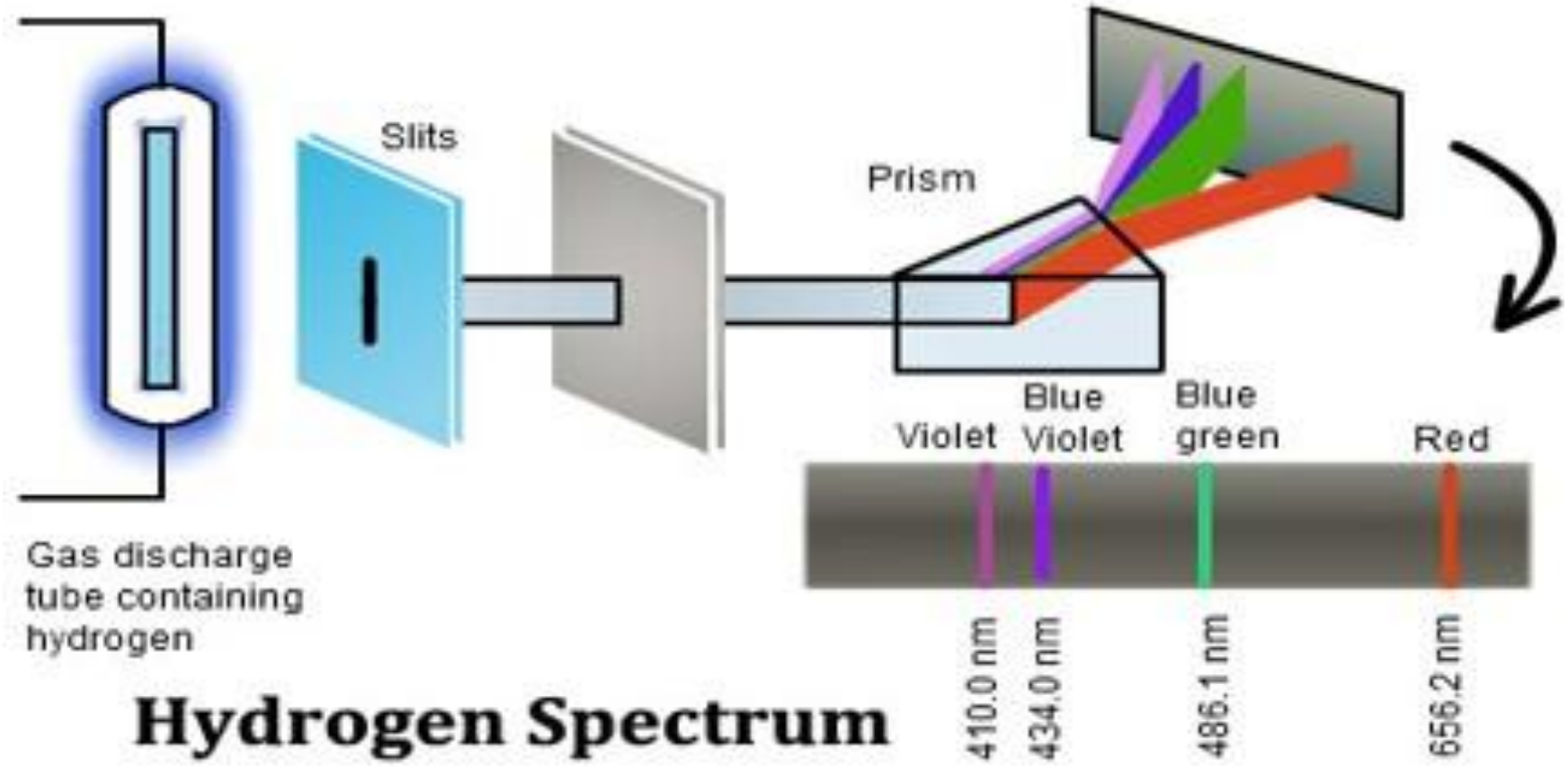
- Emission spectrum of atoms
- Periodic chemical properties of elements

Spherical coordinates



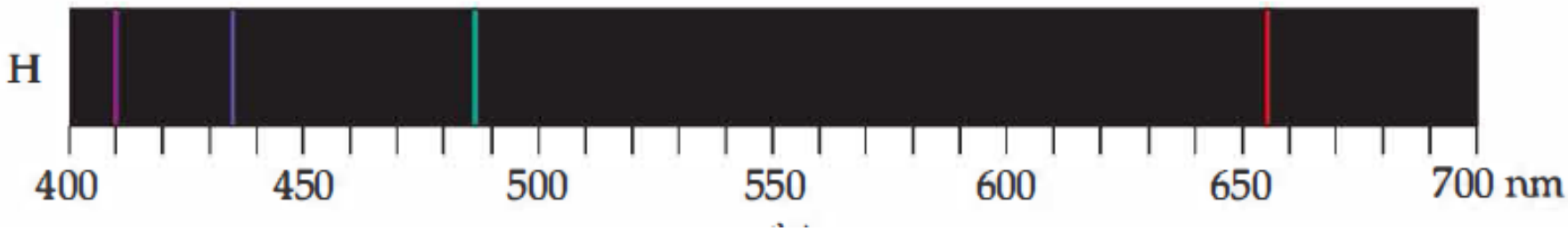
Electron is dimensionless particle (equal to **point** in math)

Emission spectrum of Hydrogen



https://assignmentpoint.com/introduction-of-hydrogen-spectrum/

Emission spectrum of Hydrogen



Rydberg (1888) found an empirical equation to calculate the wavelengths of the spectral lines of H

$$\frac{1}{\lambda} = (R_H)\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

λ is the wavelength of a spectral line (m),

R_H is the Rydberg constant (1 .096776 x 10^7 m $^{-1}$),

n_1 and n_2 are positive integers, **with $n_2 > n_1$**

WHY IS IT SO ???
BOHR'S MODEL (1913)

The work of Planck and Einstein paved the way for understanding how electrons are arranged in atoms

Bohr assumed that the prevailing laws of physics were inadequate to describe all aspects of atoms

To explain the line spectrum of hydrogen, Bohr relied on several assumptions

1. the electron moves around the proton in circular orbits

2. Only orbits of certain radii, corresponding to certain definite energies, are permitted

3. An electron in a permitted orbit has a specific energy and is in an "allowed“ energy state

4. An electron in an allowed energy state will not radiate energy and therefore will not spiral into the nucleus (as predicted by classical physics)

5. Energy is emitted or absorbed by the electron only as the electron changes from one allowed energy state to another

Calculations by Bohr show that the energies of the different allowed states are :

$$E = (-hcR_H)\left(\frac{1}{n^2}\right) = (-2.18 \times 10^{-18}\ \text{J})\left(\frac{1}{n^2}\right)$$

h - Planck's constant
c - the speed of light
R_H – Rydberg's constant

Energies are negative since these are bound states: positive energy must be given to extract the electron

Only the specific frequencies of light that have an energy (hν) = energy difference between states can be absorbed or emitted by the atom

From Bohr's model :

$$\Delta E = E_f - E_i = E_{\text{photon}} = h\nu$$

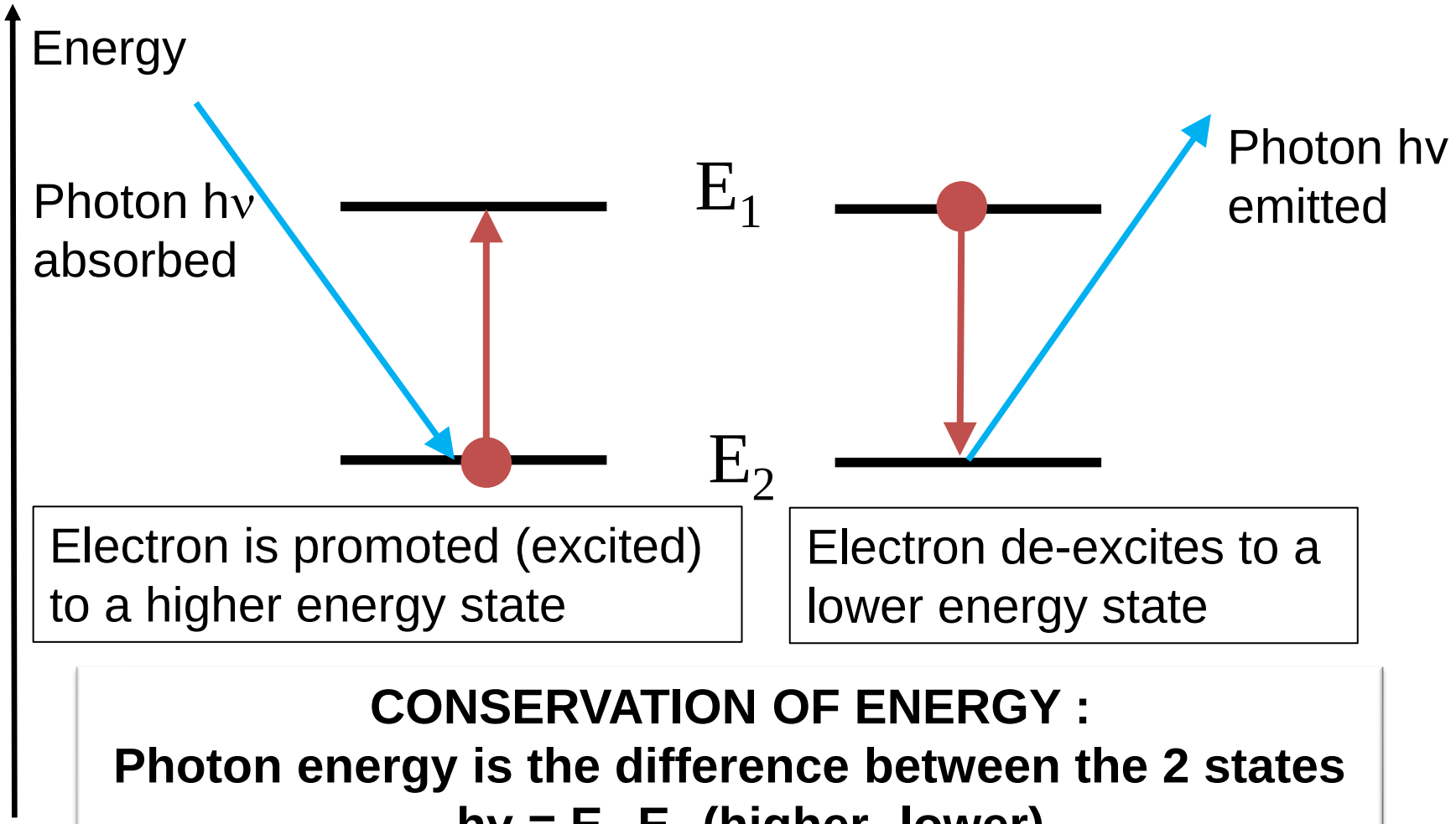
$$\Delta E = h\nu = \frac{hc}{\lambda} = (-2.18 \times 10^{-18}\ \text{J})\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda} = \frac{-hcR_H}{hc}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) = R_H\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

this equation corresponds to Rydberg's equation, which was obtained using experimental data

ELECTRON CAN CHANGE FROM ONE ALLOWED STATE TO ANOTHER ONE

- **To increase energy : absorbs a photon**
- **When decreasing energy : emits a photon**



Electron is promoted (excited) to a higher energy state

Electron de-excites to a lower energy state

CONSERVATION OF ENERGY :
Photon energy is the difference between the 2 states
$h\nu = E_1 - E_2$ (higher- lower)

In the ideal conditions (without energy loss), If a compound X absorbs the photon λ_1, it will emit photon λ_2 where:

A) $\lambda_1 = \lambda_2$

B) $\lambda_1 > \lambda_2$

C) $\lambda_1 < \lambda_2$

In the real conditions (with energy loss), If a compound X absorbs the photon λ_1, it will emit photon λ_2 where:

A) $\lambda_1 = \lambda_2$

B) $\lambda_1 > \lambda_2$

C) $\lambda_1 < \lambda_2$

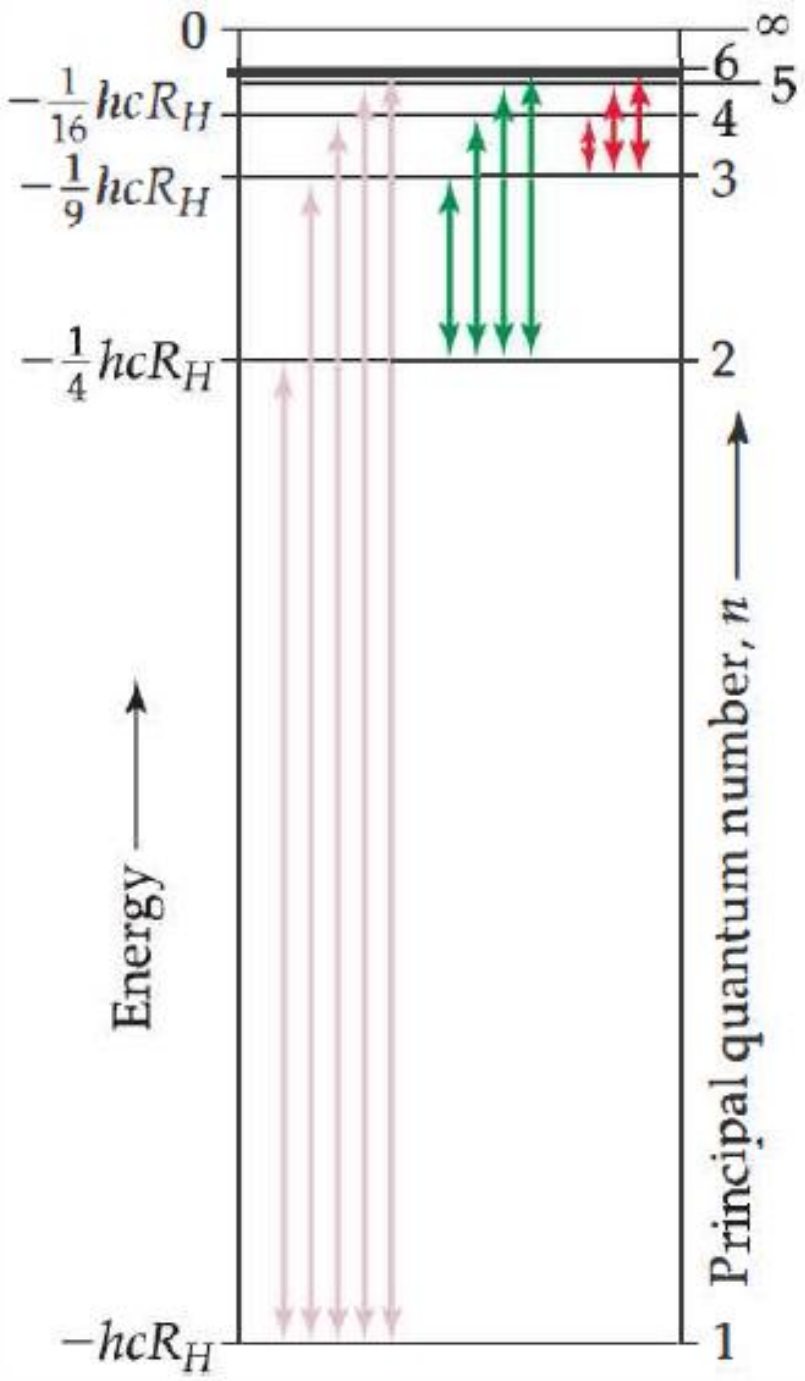
There are many possible transitions between allowed states but always $h\nu = E_1 - E_2$ (higher –lower)

Line spectrum

n is called quantum number

When n = ∞ , the energy is 0

Then the atom has lost its electron : ionized state H^+



BOHR'S MODEL : CONCLUSIONS

GREAT SUCCESS ! Bohr's model explains the line spectrum of the hydrogen atom

BUT : it cannot explain the spectra of other atoms

Two important useful ideas Bohr's model:

(1) Electrons in atoms have only certain discrete energy levels

(2) Energy is exchanged in moving an electron from one level to another one

The science of Fireworks



Pls describe the science occurred in Fireworks?

Aerial fireworks

1. **Stick ("tail")**: make sure the firework goes in right direction
2. **Fuse: Where you initiate the operation (“burn”)**
3. **Charge ("motor")**: A relatively crude explosive designed to blast a firework up into the sky
4. **Effect**: Provide amazing colors and structure (with multiple components)
5. **Head:** "nose cone" to make the firework faster and more aerodynamic and improve the chance of it going in a straight line

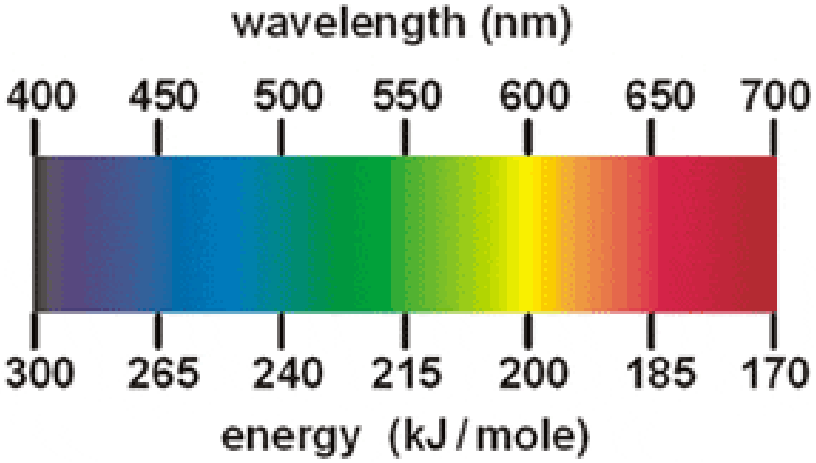


3. Charge (Motor):

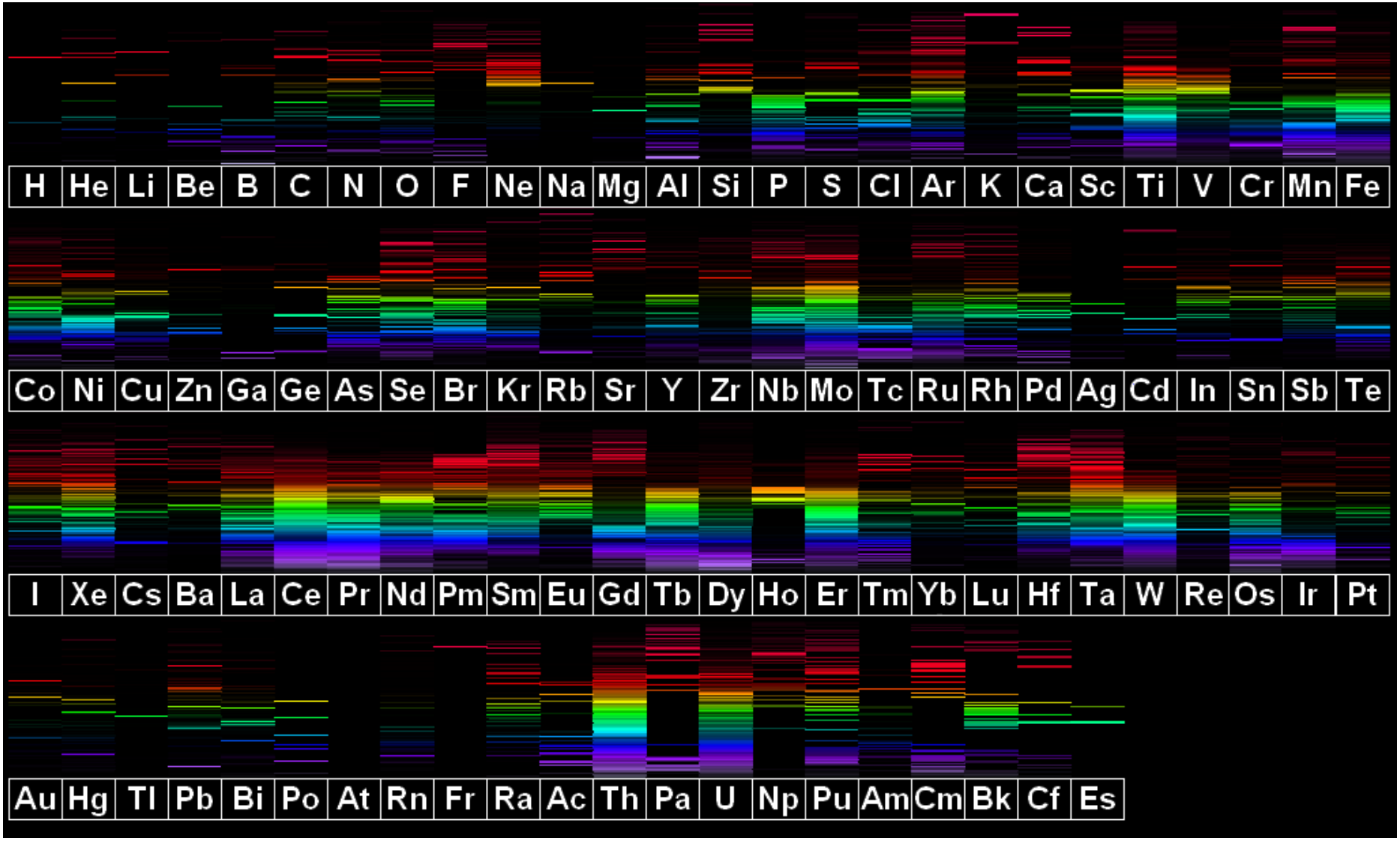
$$2KNO_3 + S + 3C \rightarrow K_2S + N_2 + 3CO_2 + \Delta G^0$$

4. Effect: Generating Color

Na compounds	: Yellow/ Organe
K	: Violet
Ba	: Green/ Blue
Ca	: Red
Sr:	: Red



Visible spectra of Elements

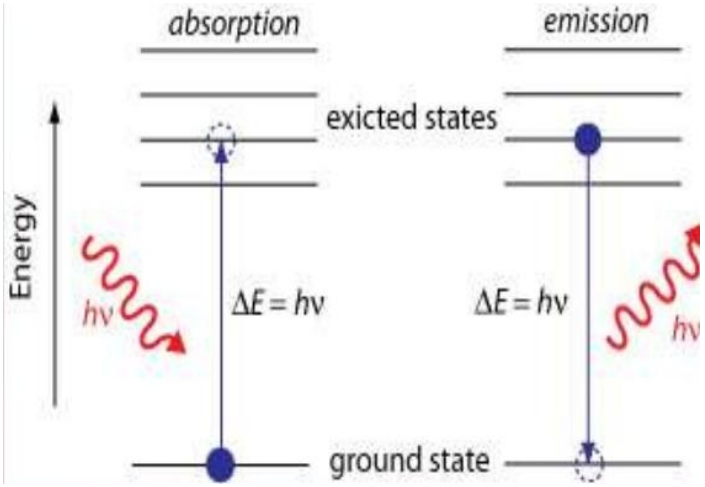


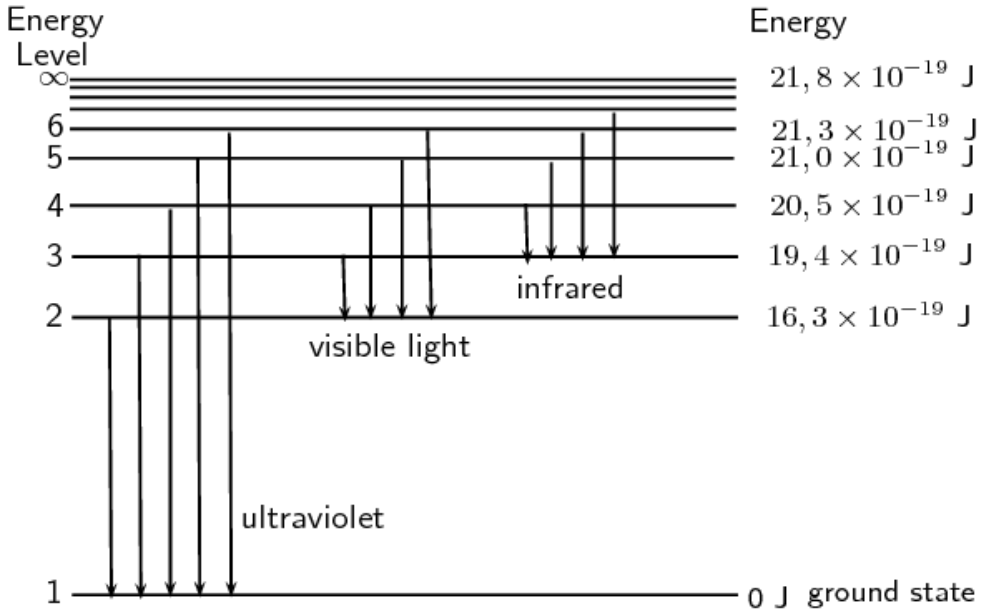
Atomization

Atomization

Na^+Cl^-. Solvent $\rightarrow Na^+Cl^- \rightarrow Na_{(at)}\,Cl_{(at)} \rightarrow Na_{(at)}^* \rightarrow Na_{(at)}$ + light emission

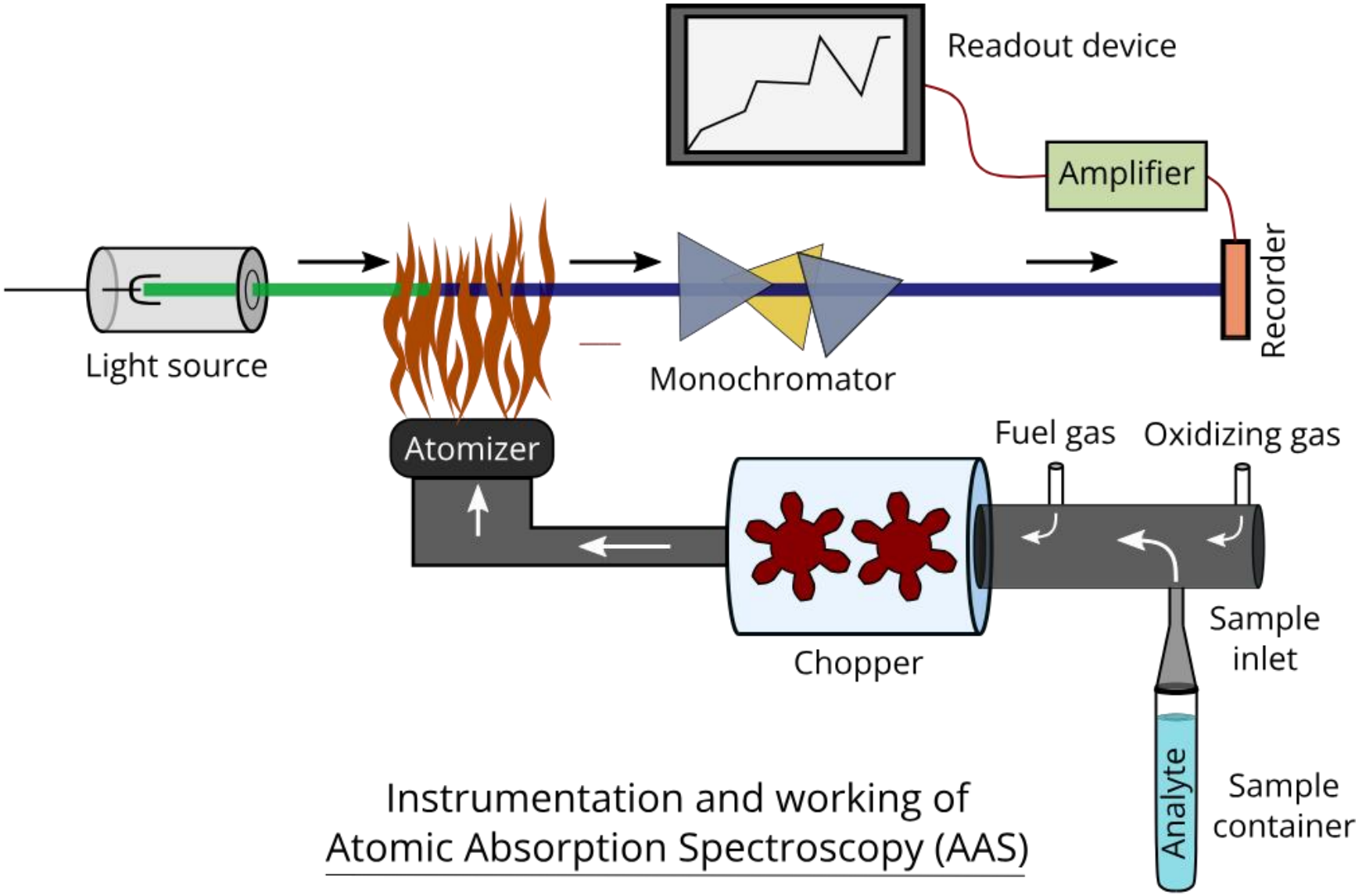
Principles



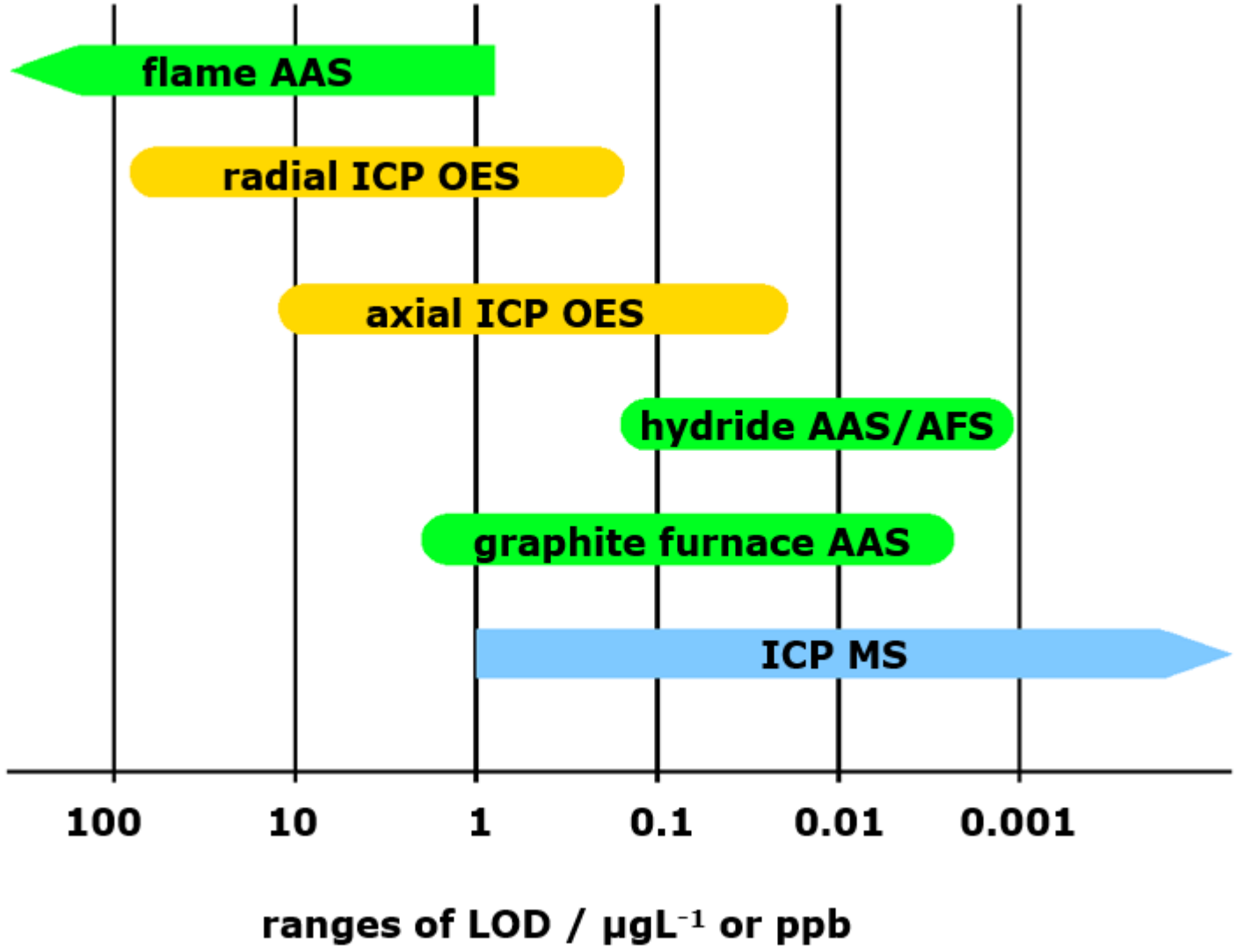


Not just for the party…

But also for work (hard work)….



Instrumentation and working of
Atomic Absorption Spectroscopy (AAS)


flame AAS
radial ICP OES
axial ICP OES
hydride AAS/AFS
graphite furnace AAS
ICP MS
100
10
1
0.1
0.01
0.001
ranges of LOD / µgL-1 or ppb

5. QUANTUM PHYSICS.
THE WAVE BEHAVIOR OF MATTER

Bohr could not explain why electrons were restricted to some fixed distances around the nucleus…

Louis de Broglie (1924) proposed that if light can behave as a particle (photon) then

PARTICLES (electrons and others) MAY EXHIBIT WAVE CHARACTERISTICS.

Another revolutionary idea !!

De Broglie : PARTICLE MOVEMENT AND WAVE PROPERTIES ARE RELATED BY

$$\lambda = \frac{h}{mu}$$

where

λ = wavelength “associated” to the particle

m = mass (kg)

u = velocity (m/s)

for an electron moving at 5.0×10^{6} m/s $\lambda = 1.46\times10^{-10}$ m (0.146 nm). Quite small ! So not detected earlier…

The wave properties of matter are proved by a number of experiments ! And are used every day in some electronic devices (‘tunnel diodes’ in your mobile phone)

$$\lambda = \frac{h}{mu}$$

Calculate the wavelength associated to your movement

$h = 6.62607004 \times 10^{-34}\ m^2\ kg\ /\ s$

QUANTUM MECHANICS : THE PHYSICS OF MATTER WAVES

De Broglies's waves of matter brought some justification to Bohr's model but

Heisenberg, Schrödinger and others improved and generalised these ideas and very fast developed a new physics

Schrödinger's equation allows calculating these waves (wavefunctions) in systems like atoms

MAIN FEATURES of QUANTUM MECHANICS

A wave cannot be precisely localized …

As electrons behave like waves they cannot be precisely localized

More precisely we cannot know simultaneously their speed and their position

(uncertainty principle)

Electrons are described by wave functions which give their probability of presence in some portion of space and their energy

JUST TO FRIGHTEN YOU !

Here is Schrödinger's equation !

$$\frac{-\hbar^2}{2m}\nabla^2\psi(x,y,z,t)+V(x,y,z,t)\psi(x,y,z,t)=i\hbar\frac{\partial}{\partial t}\psi(x,y,z,t)$$

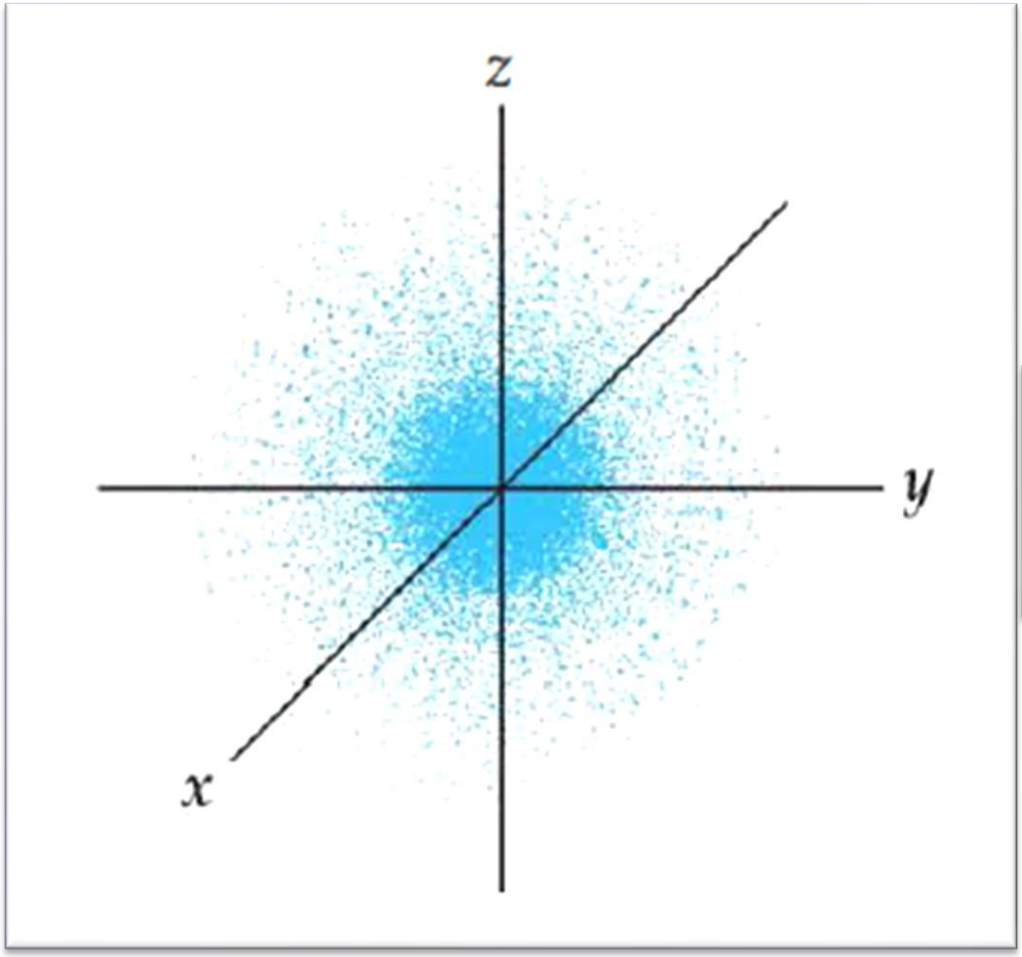
where

$$\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}=\nabla^2$$

In some way this replaces Newton's law F = ma

Solving Schrodinger's equation for an atom leads to a series of functions called wave functions, Ψ (psi) that describe the electron in an atom

The wave function ψ(x,y,z) will give the probability density or the electron density (ψ^2) = probability that the electron will be in a certain region of space



Example of an electron density distribution

the density of dots represents the probability of finding the electron.

6. ATOMIC ORBITALS

The wavefunctions are stationary waves around the nucleus.

The solution to Schrodinger's equation for the hydrogen atom yields a set of wave functions and corresponding energies

There is a lowest-energy stationary wave, and higher energy ones.

These wave functions are called ORBITALS

Each orbital describes a specific distribution of electron density in space

Each orbital has a characteristic energy and shape

Each atomic orbital in an atom is characterised by a unique set of three quantum numbers (from Schrödinger's wave equation) :

$$n, \ell, \text{ and } m_\ell$$

Quantum numbers

n = principal quantum number

has positive integer values

n = 1,2,3,…

If n increases,

- **the orbital becomes larger**
- **the electron spends more time farther from the nucleus**
- **the electron has a higher energy**
- **the electron is less tightly bound to the nucleus**

ℓ = angular momentum quantum number, has integer values from 0 to (n - 1) for each value of n

(for instance if n=1 ℓ =0 ; if n=3 ℓ can be 0, 1, 2)

ℓ defines the shape of the orbital

Orbitals are named s, p, d, f depending on the value of ℓ

Value of l	0	1	2	3
Orbital name	s	p	d	f

sharp
principal
diffuse
fundamental

m_ℓ = **magnetic quantum number**

describes the orientation of the orbital in space

has integer values between

-ℓ and ℓ, including 0

How to manage these quantum numbers ?

$n = 1, 2, 3\ldots \quad 0 \leq \ell \leq n-1 \quad -\ell \leq m_l \leq +\ell$

n=1 — ℓ : only possibility $\ell=0$

$\ell=0$ so $m_l=0$. Just 1 orbital (1;0;0)

Orbital 1s because n=1 and s means l=0

n=2 — then $\ell=0$ or $\ell=1$

if $\ell=0$ $m_l=0$: 1 orbital (2;0;0) — Orbital 2s (n=2 and l=0)

if $\ell=1$ $m_l = -1$ or 0 or +1 : 3 orbitals (2,1,-1), (2,1,0), (2,1,1)

3 Orbitals 2p (n=3 and p means l=1)

n=3 then ℓ=0 or ℓ=1 or ℓ=2

if ℓ=0 m_l=0 : 1 orbital Orbital 3s (n=3 and l=0)

if ℓ=1 m_l = -1 or 0 or +1 3 Orbitals 3p (n=3 and p means l=1)

if ℓ=2 m_l =-2 or -1 or 0 or +1 or +2

5 Orbitals 3d (n=3 and d means l=2)

n=4 ℓ=0 or 1 or 2 or 3

1 orbital 4s (l=0)

3 orbitals 4p (l=1)

5 orbitals 4d (l=2)

7 orbitals 4f (l=3 so m_l = -3…0 …+3 7 possible values)

Summary table

TABLE 6.2 ■ Relationship among Values of n, l, and m_l through $n = 4$

n	Possible Values of l	Subshell Designation	Possible Values of m_l	Number of Orbitals in Subshell	Total Number of Orbitals in Shell
1	0	1*s*	0	1	1
2	0	2*s*	0	1	
	1	2*p*	1, 0, −1	3	4
3	0	3*s*	0	1	
	1	3*p*	1, 0, −1	3	
	2	3*d*	2, 1, 0, −1, −2	5	9
4	0	4*s*	0	1	
	1	4*p*	1, 0, −1	3	
	2	4*d*	2, 1, 0, −1, −2	5	
	3	4*f*	3, 2, 1, 0, −1, −2, −3	7	16

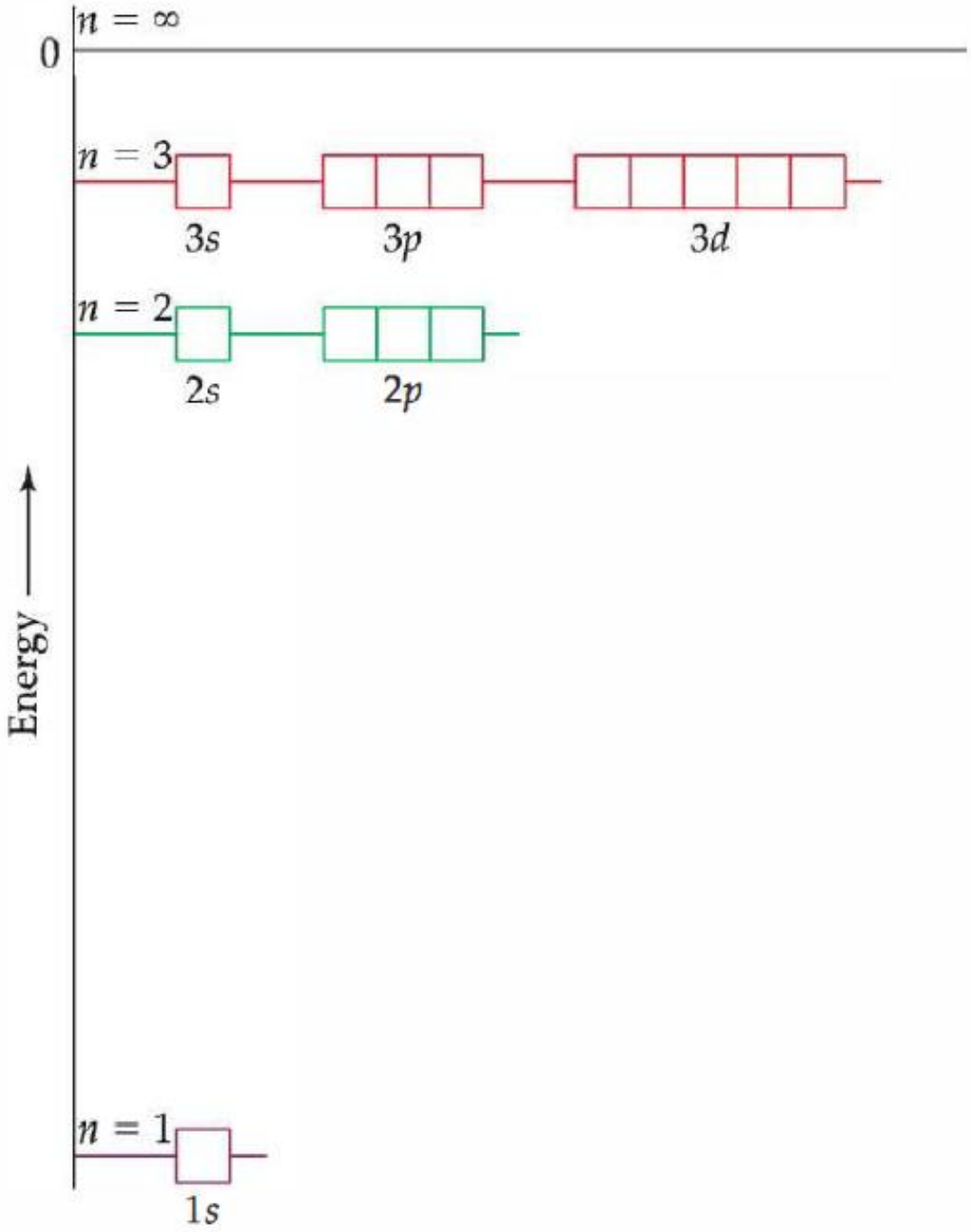
- orbitals with the same value of n form a "shell".

- orbitals that have n = 3, for example, are said to be in the 3rd shell.

- orbitals with the same n and ℓ values form a "subshell"

- in shell number n there are n subshells

- the total number of orbitals in a shell is n^2



Orbital energy levels in the H atom.

Energy depends **only on n**.

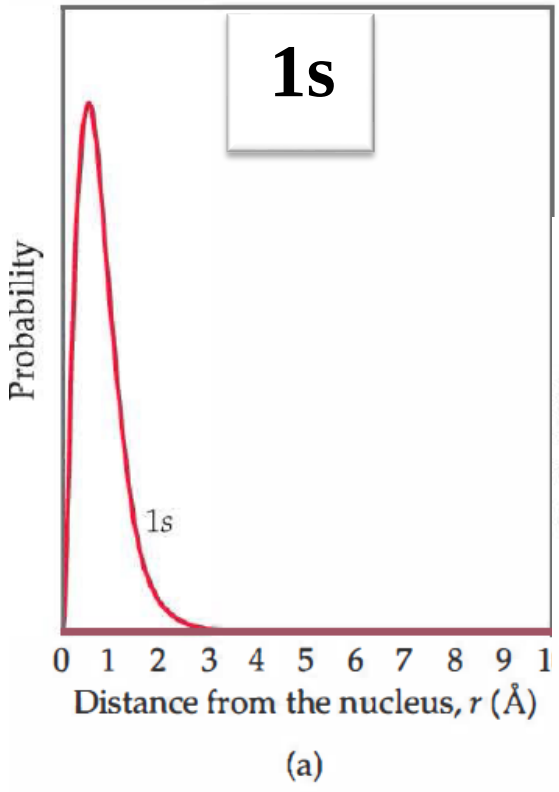
All the orbitals in one shell have the same energy

REPRESENTATIONS OF ORBITALS

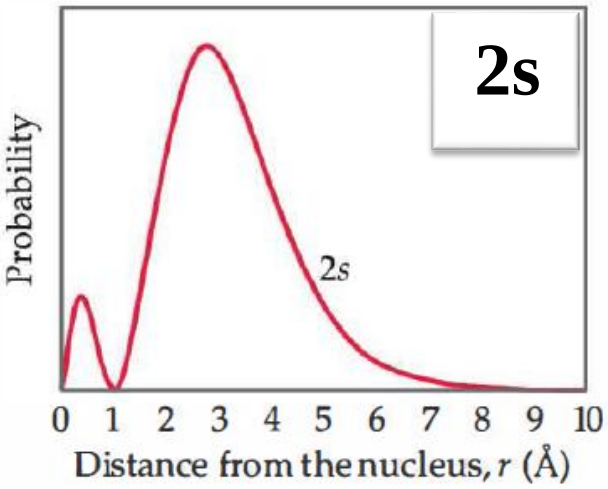
1) probability= f(distance to nucleus)

example : s orbitals

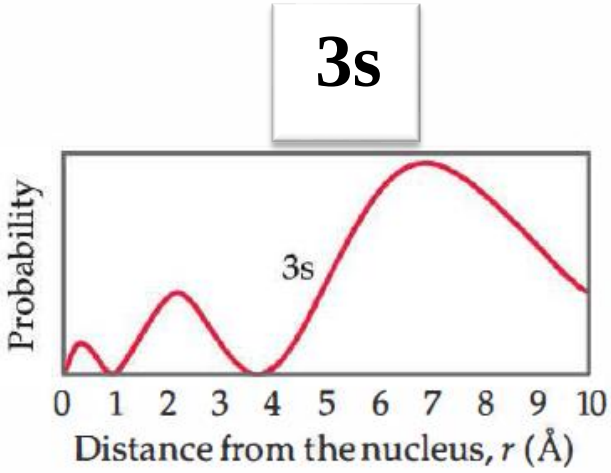
As n increases, the most likely distance from the nucleus to find the electron increases



(a)



(b)

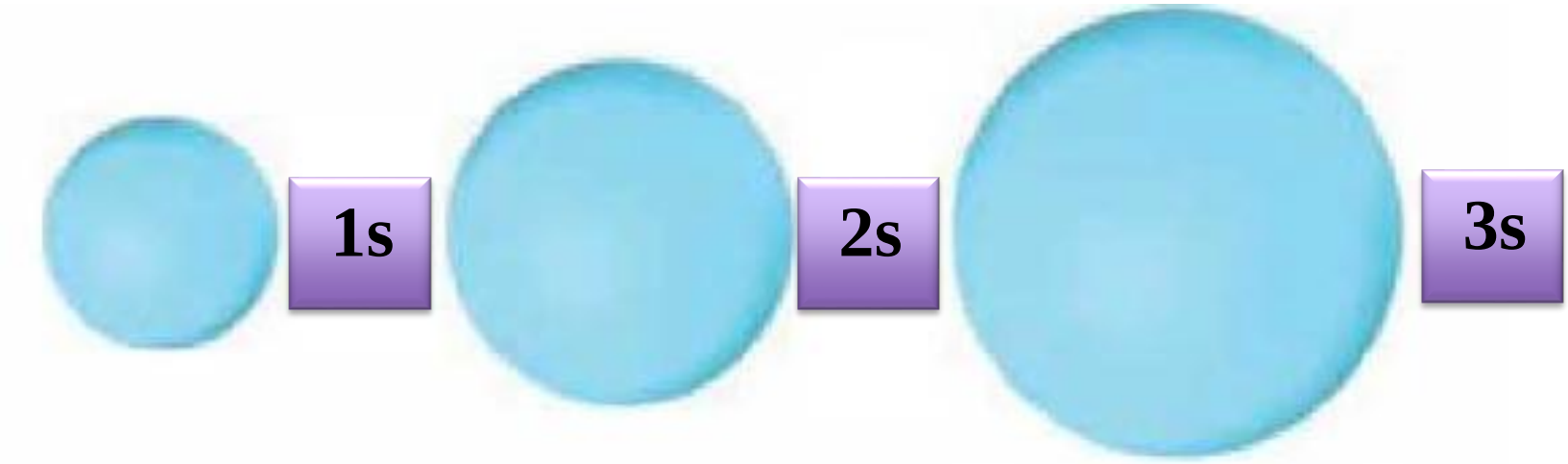


(c)

REPRESENTATIONS OF ORBITALS
2) volumes within which 90% chances to find the electron

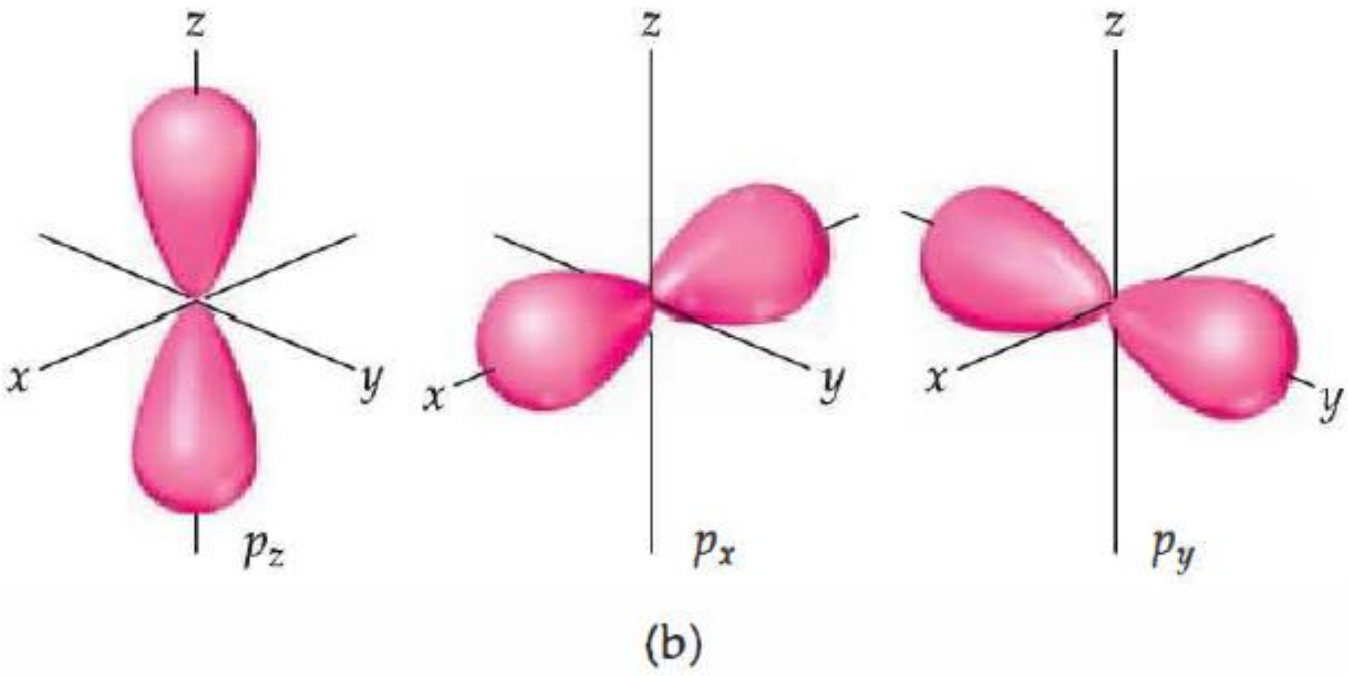
90% is arbitrary. We can chose 80 or 92 or 95%.... But not 100 % !

s orbitals : spherical



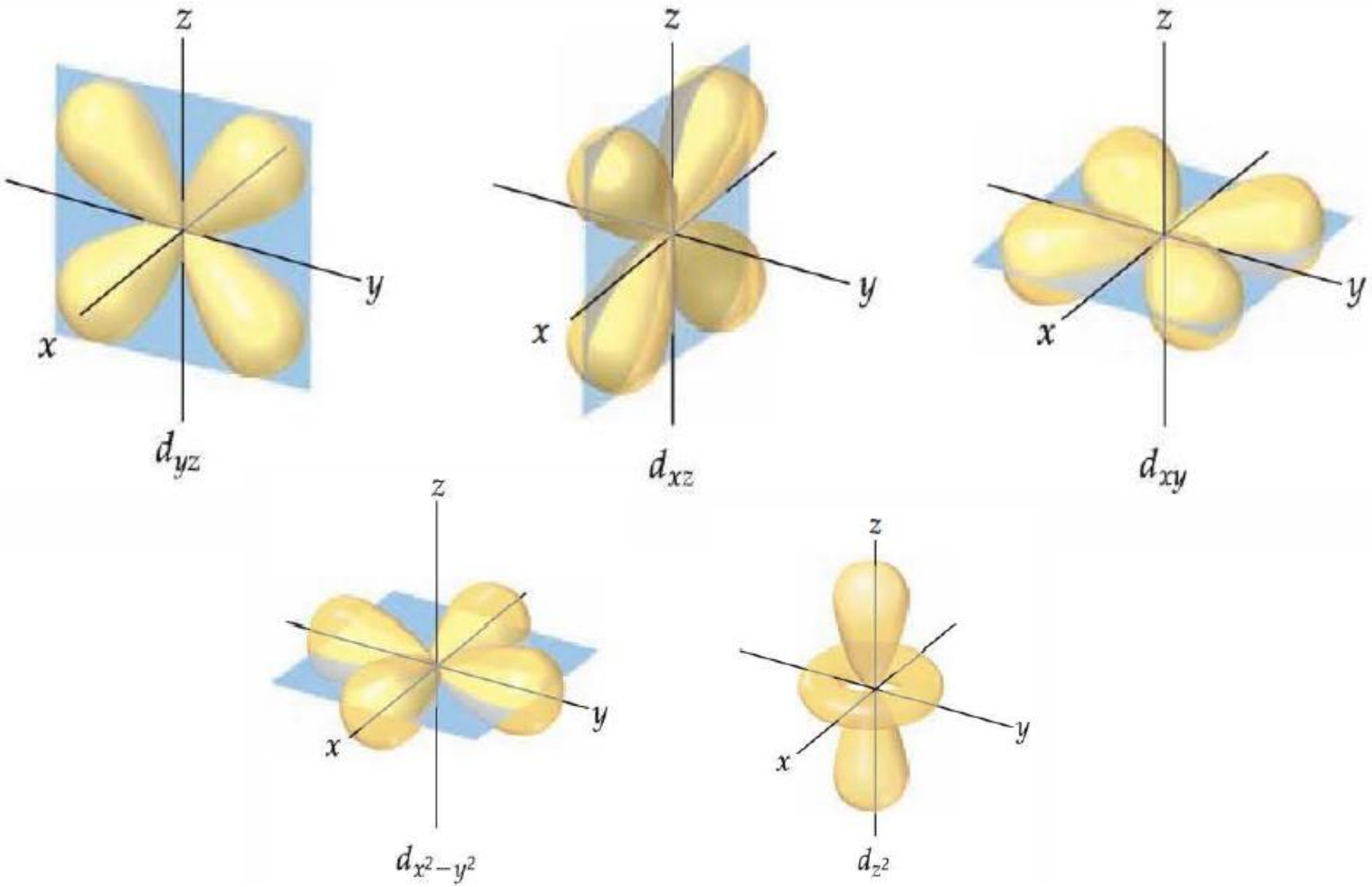
boundary surfaces for p orbitals (dumbbell-shaped)

p orbitals have preferential orientations



p_z (m=0)	p_x (m=1)	p_y (m=-1)

boundary surfaces for d orbitals



The shapes of the f orbitals are even more complicated …..

7. MANY-ELECTRON ATOMS

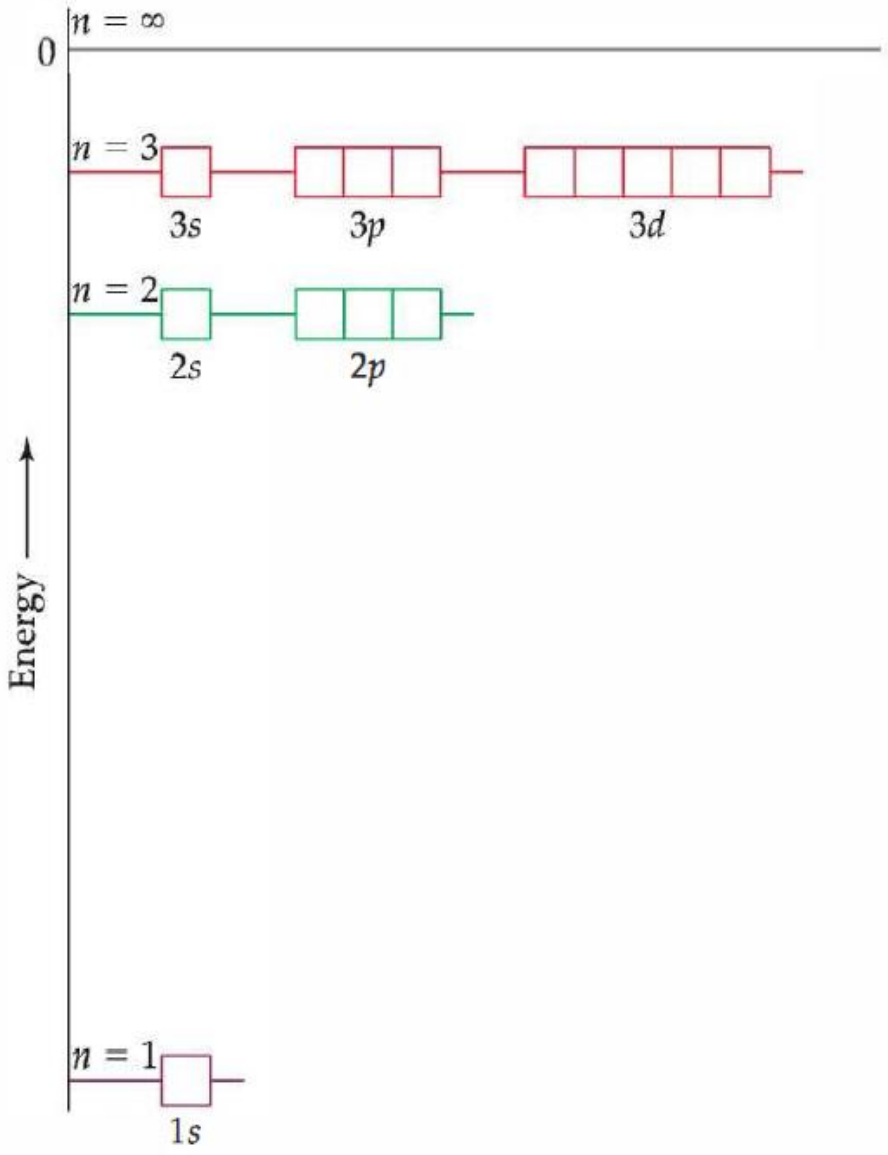
Difference with H : in many-electron atoms, for a given value of n, the energy of an orbital increases with increasing value of ℓ.

2s < 2p

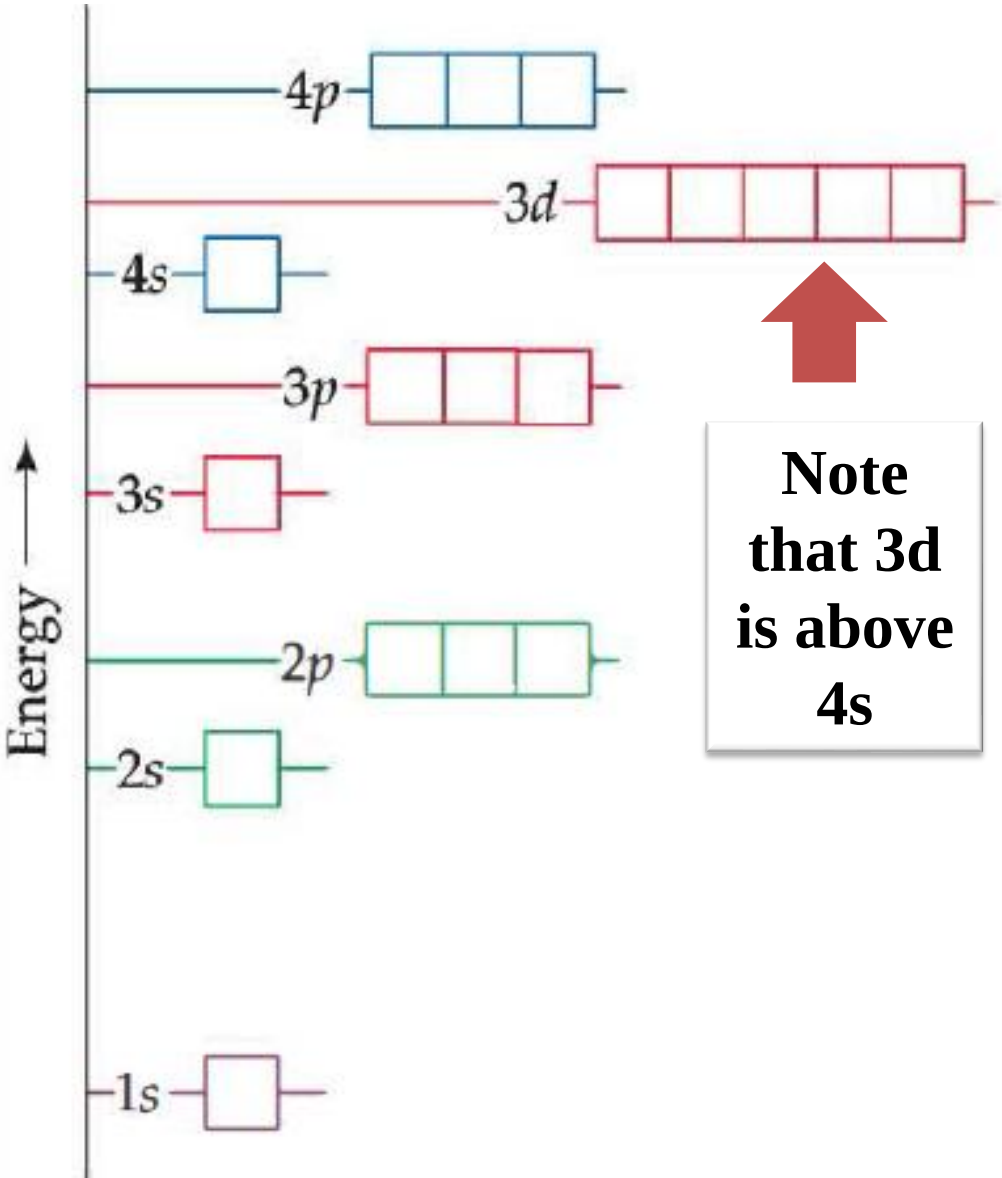
3s < 3p <3d

etc.

Orbital energy levels in H or one electron ions



Orbital energy levels In many electron atoms

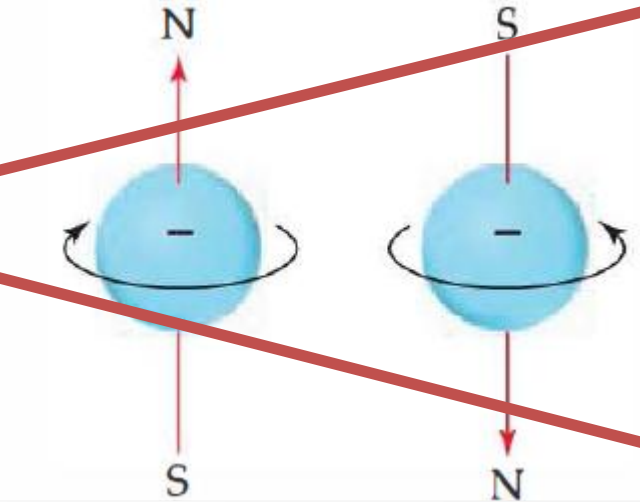


Note that 3d is above 4s

A little more physics : electron Spin

Electrons (and other particles) have an intrinsic property, called spin, that causes each electron to behave as a tiny magnet.

People say : as if it were a tiny sphere spinning around one axis



BUT :

AN ELECTRON IS NOT A SPHERE (it is dimensionless) and IT DOES NOT SPIN AROUND AN AXIS

Spin is quantified : 2 possible values

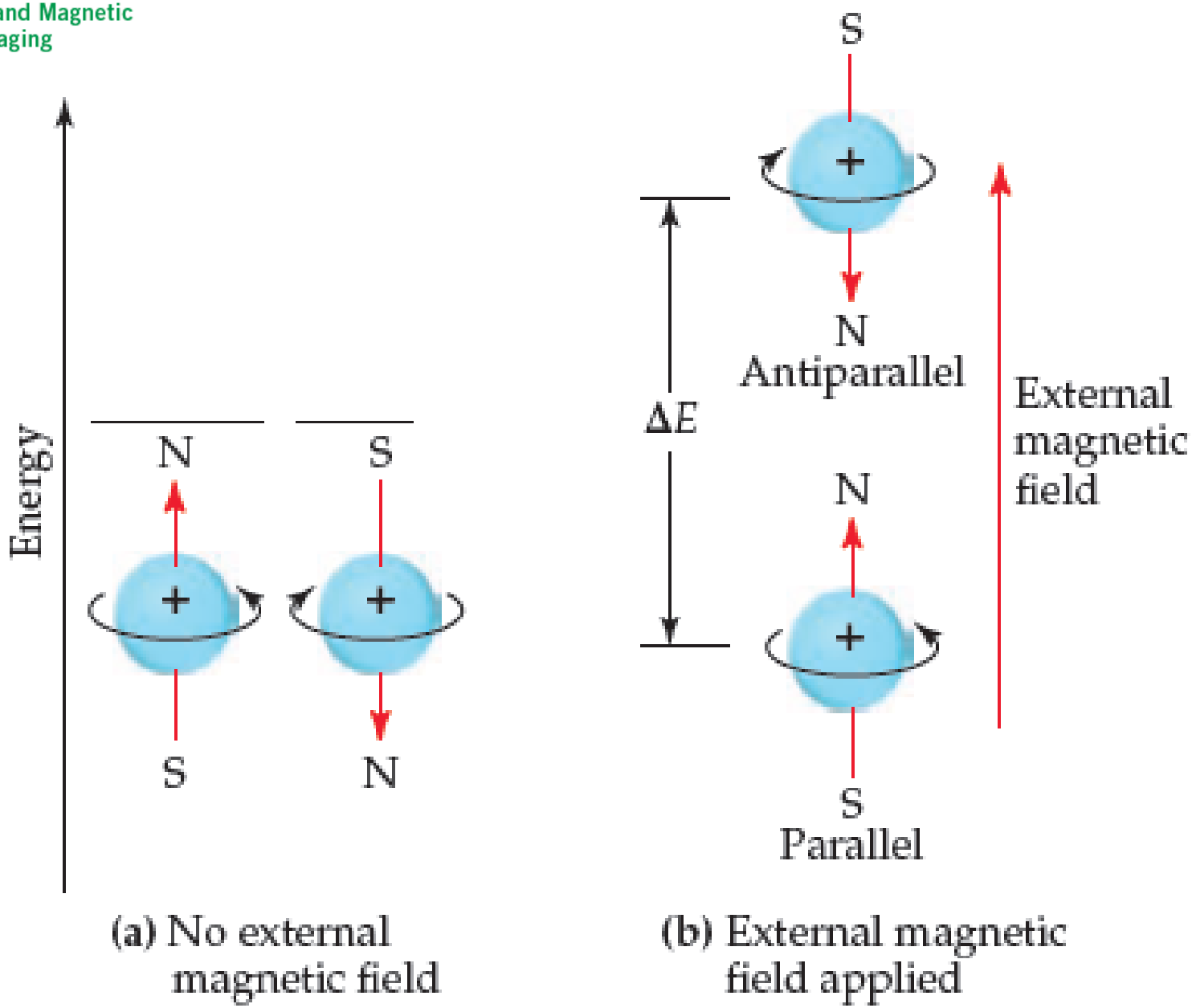
4th quantum number m_s = -1/2 or +1/2

Particles with spin obey Pauli exclusion principle

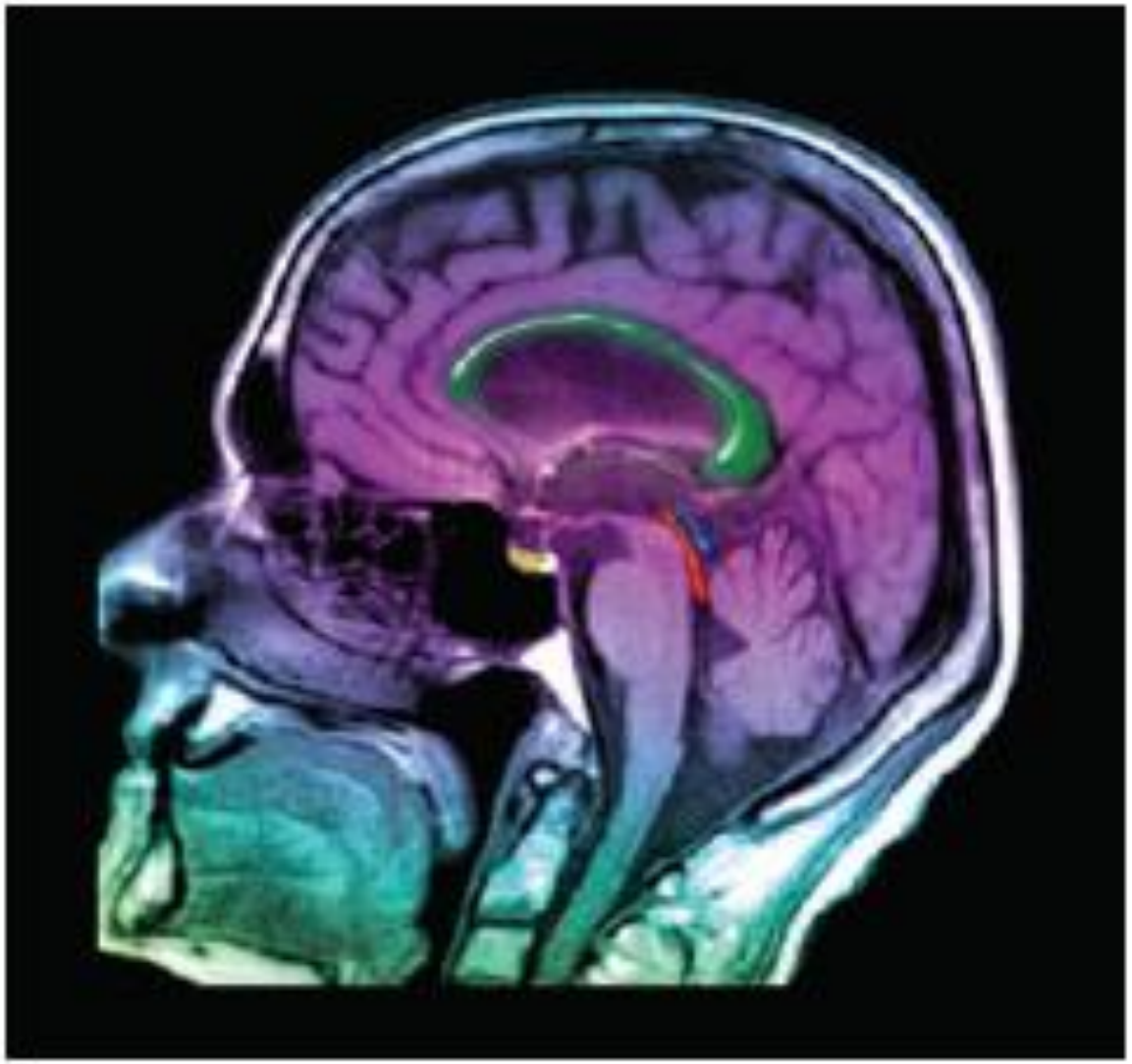
No two electrons in an atom can have the same set of four quantum numbers n, ℓ, m_ℓ and m_s



One orbital (defined by n,ℓ, m_ℓ) can hold a maximum of two electrons with opposite spins -1/2 and +1/2



▲ **Figure 6.27** **Nuclear spin.** Like electron spin, nuclear spin generates



▲ **Figure 6.28** **MRI image.** This image of a human head, obtained using magnetic resonance imaging, shows a normal brain, airways, and facial tissues.

Spin is quantified : 2 possible values

4th quantum number m_s = -1/2 or +1/2

Particles with spin obey Pauli exclusion principle

No two electrons in an atom can have the same set of four quantum numbers n, ℓ, m_ℓ, and m_s

One orbital (defined by n,ℓ, m_ℓ) can hold a maximum of two electrons with opposite spins -1/2 and +1/2

For the table that follows, write which orbital goes with the quantum numbers. Don't worry about *x*, *y*, *z* subscripts. If the quantum numbers are not allowed, write "not allowed."

N	l	m_l	Orbital
2	1	−1	2*p* (example)
1	0	0	
3	−3	2	
3	2	−2	
2	0	−1	
0	0	0	
4	2	1	
5	3	0	

Determine whether each of the following sets of quantum numbers for the hydrogen atom are valid. If a set is not valid, indicate which of the quantum numbers have a value that is not valid.

(**a**) n=4, l=1, m_l = 2, m_s = -1/2

(**b**) n= 4, l= 3, m_l = -3, m_s = + 1/2

(**c**) n = 3, l = 2, m_l = -1, m_s = +1/2

(**d**) n = 5, l = 0, m_l = 0, m_s = 0

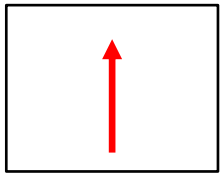
(**e**) n = 2, l = 2, m_l= 1, m_s= +1/2

7. MANY-ELECTRON ATOMS (cont.)

A convenient scheme for orbitals and electrons

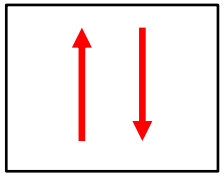
1 Orbital = 1 box

Electrons =arrows representing spin (1/2 or -1/2 , up or down)



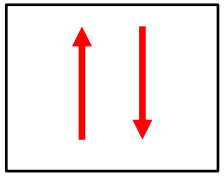
Orbital 1s with 1 electron

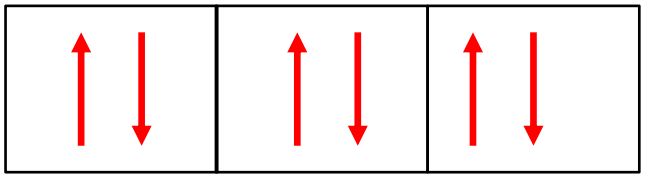
unpaired electron



Orbital 1s with 2 electrons

paired electrons





“full” orbitals 2s and 2p

The 3 orbitals 2p are grouped since they have the same energy

7. ELECTRON CONFIGURATIONS

Electron configuration: the way in which the electrons are distributed among the various orbitals of an atom

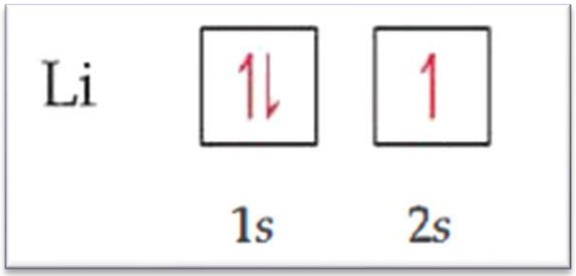
The most stable electron configuration of an atom is the ground state = the lowest possible energy state

Orbitals are "filled" in order of increasing energy, with no more than two electrons per orbital

Example of electron configuration

The lithium atom (Li) has 3 electrons.

The 1s orbital can accommodate 2 of the electrons. The third one goes into the next lowest energy orbital, the 2s.



Orbital diagram

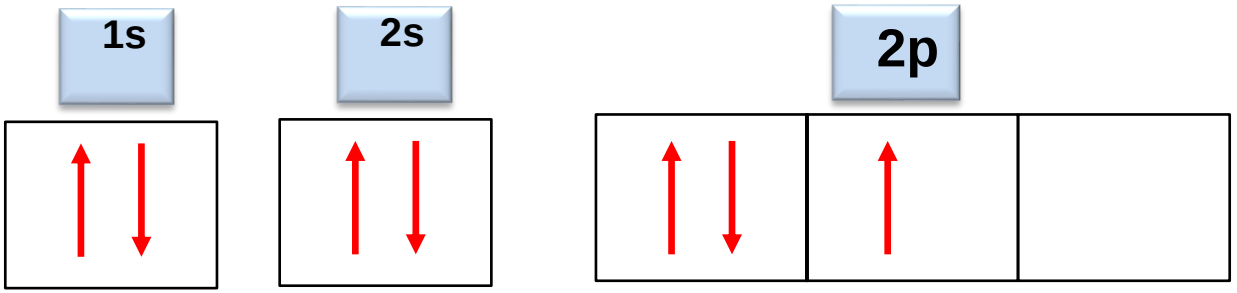
↑ **Positive spin magnetic quantum number $m_s = +1/2$**

↓ **Negative spin magnetic quantum number $m_s = -1/2$**

We write it like this :

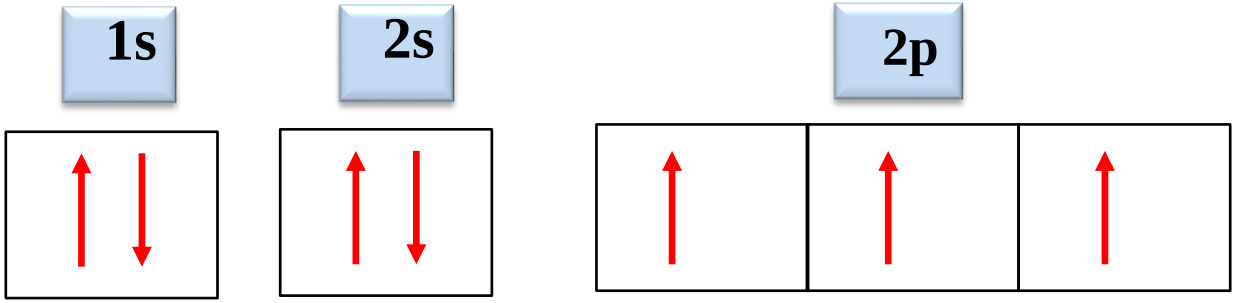
Lithium electron configuration is $1s^2 2s^1$

What about Nitrogen ? (N : 7 electrons)



??? No !!!

HUND'S RULE: for degenerate orbitals (orbitals with same energy), the lowest energy state is attained when the number of electrons with the same spin is maximized

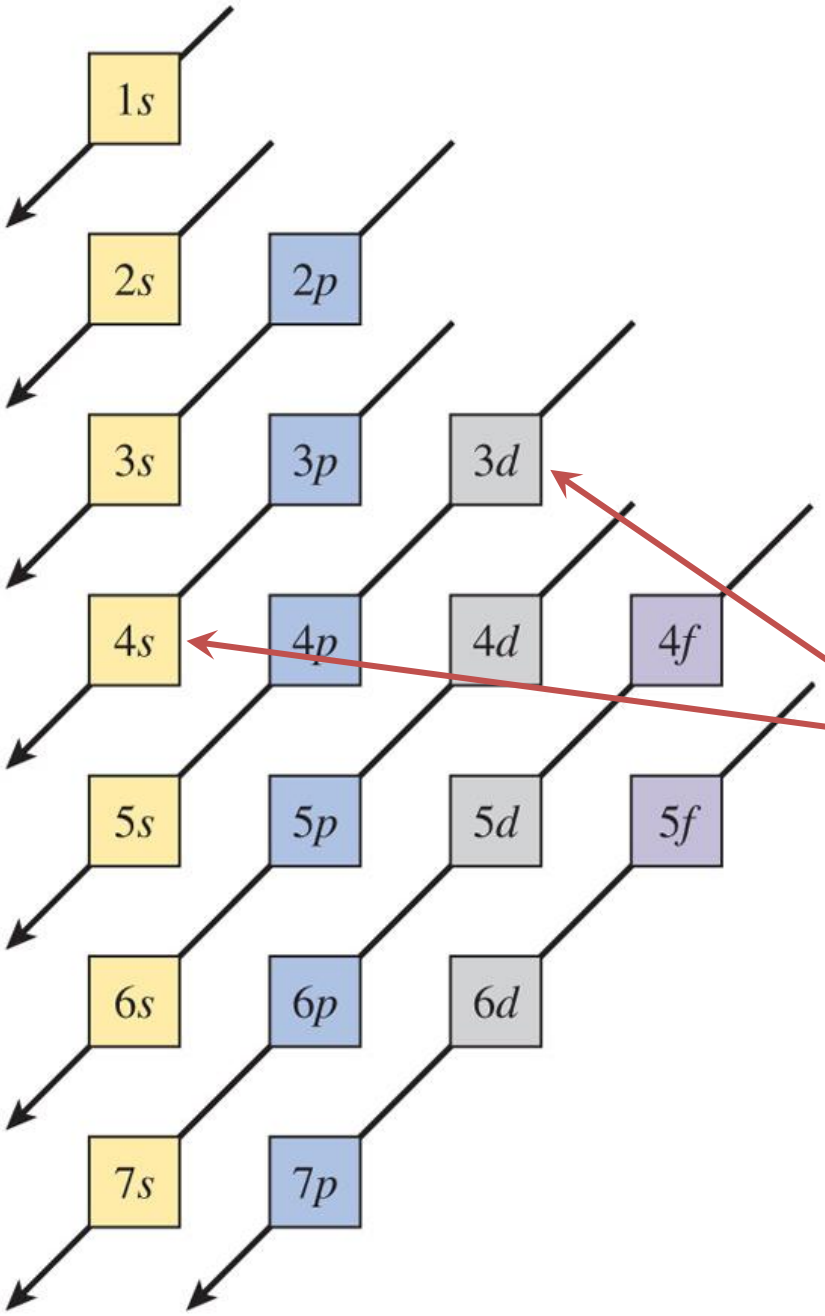


1 electron in each p orbital

N: $1s^2\ 2s^2\ 2p^3$

TABLE 6.3 ■ Electron Configurations of Several Lighter Elements

Element	Total Electrons	Orbital Diagram				Electron Configuration
		1*s*	2*s*	2*p*	3*s*	
Li	3	↿⇂	↿			$1s^2 2s^1$
Be	4	↿⇂	↿⇂			$1s^2 2s^2$
B	5	↿⇂	↿⇂	↿		$1s^2 2s^2 2p^1$
C	6	↿⇂	↿⇂	↿ ↿		$1s^2 2s^2 2p^2$
N	7	↿⇂	↿⇂	↿ ↿ ↿		$1s^2 2s^2 2p^3$
Ne	10	↿⇂	↿⇂	↿⇂ ↿⇂ ↿⇂		$1s^2 2s^2 2p^6$
Na	11	↿⇂	↿⇂	↿⇂ ↿⇂ ↿⇂	↿	$1s^2 2s^2 2p^6 3s^1$



How to build electron configuration of atoms ? The diagonal rule to fill orbitals with electrons

Remember 4s < 3d so
4s is filled before 3d
Same for 5s and 4d,4f etc.

Exceptions to the rule:

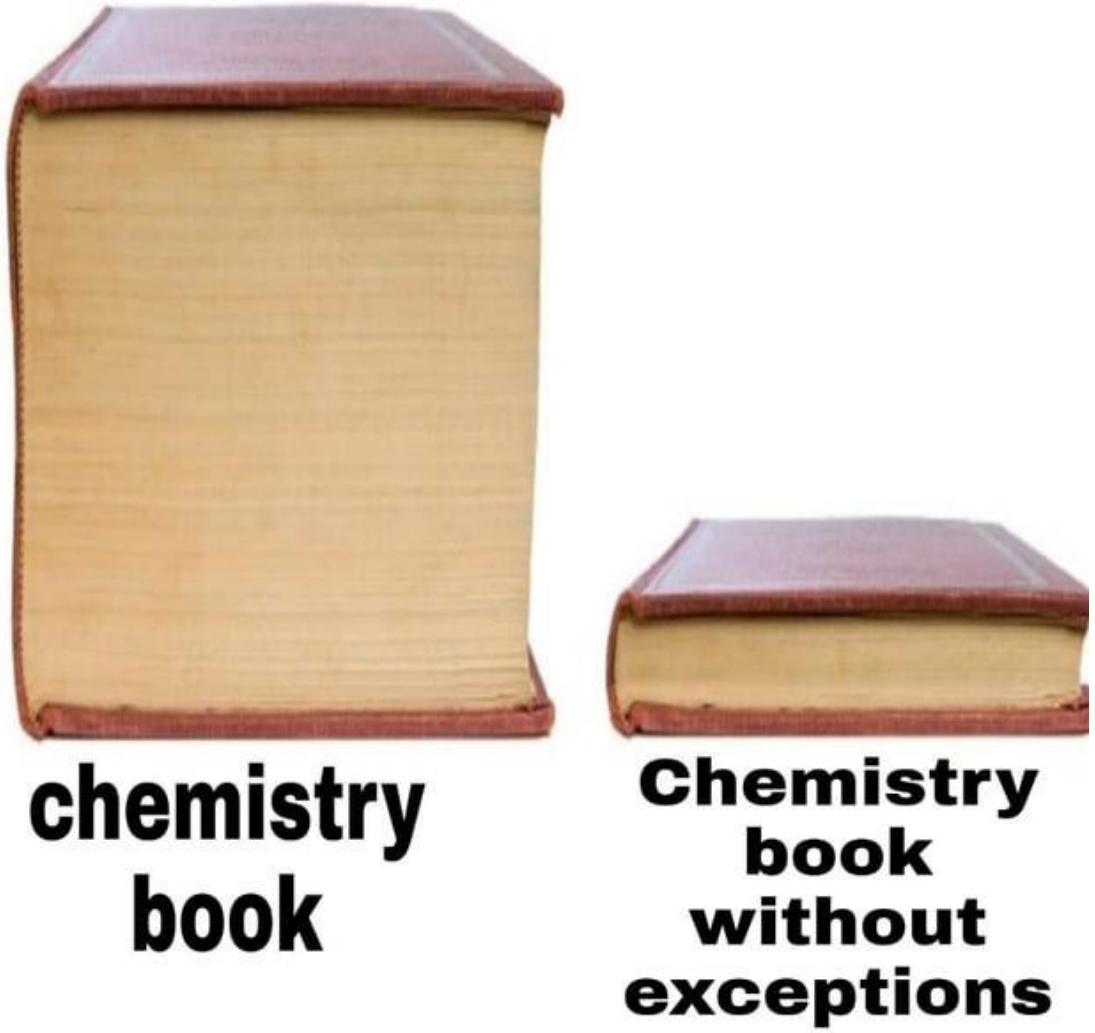
Cu (Z = 29) and Cr (Z = 24)
Cu = $4s^1 3d^{10}$ instead of $4s^2 3d^9$
Cr = $4s^1 3d^5$ instead of $4s^2 3d^4$

I hate Chemistry because of a lot EXCEPTIONS….?

I LOVE Chemistry because of a lot EXCEPTIONS….!!!

EXCEPTIONS…

Because we have not yet fully understood the Nature and its chemical rules



The following electron configurations represent excited states. Identify the elements and write its ground state condensed electron configuration.

(**a**) $1s^2 2s^2 2p^4 3s^1$

(**b**) $[Ar] 4s^1 3d^{10} 4p^2 5p^1$

(**c**) $[Kr] 5s^2 4d^2 5p^1$

The electron configuration of $_{14}Si$ is:

A) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

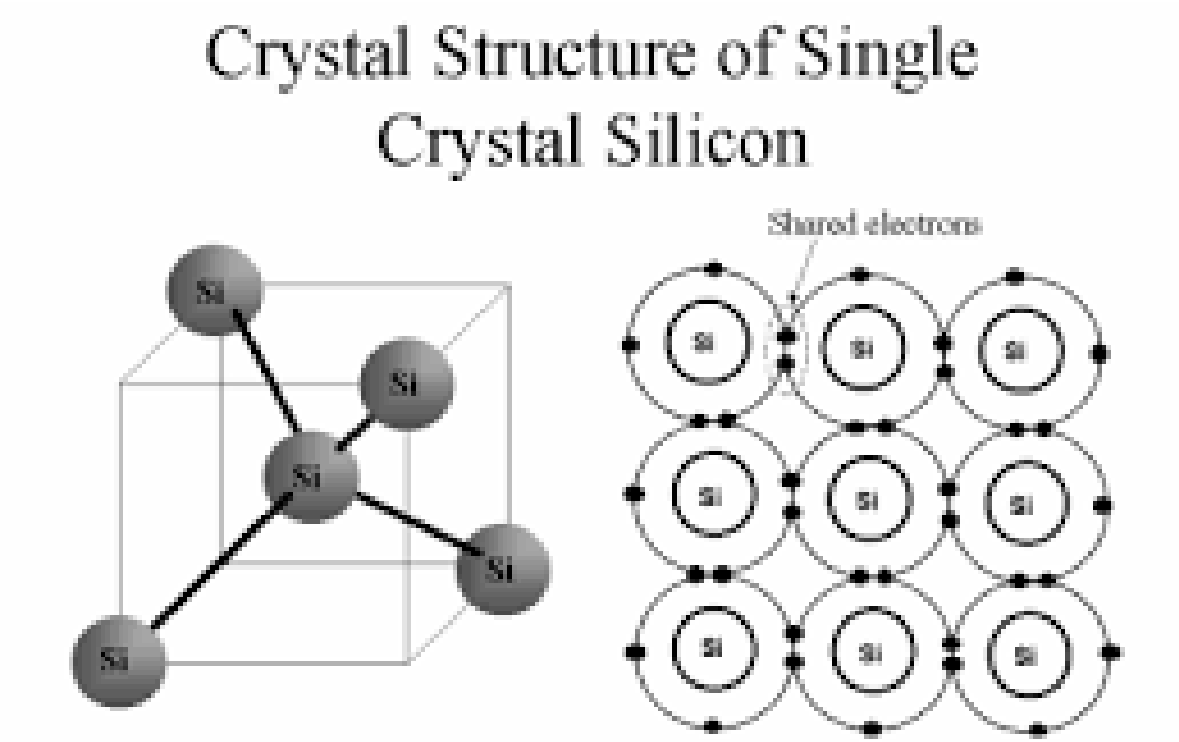
B) $1s^2\ 2s^1\ 2p^6\ 3s^1\ 3p^3$

The electron configuration of $_{5}B$ is:

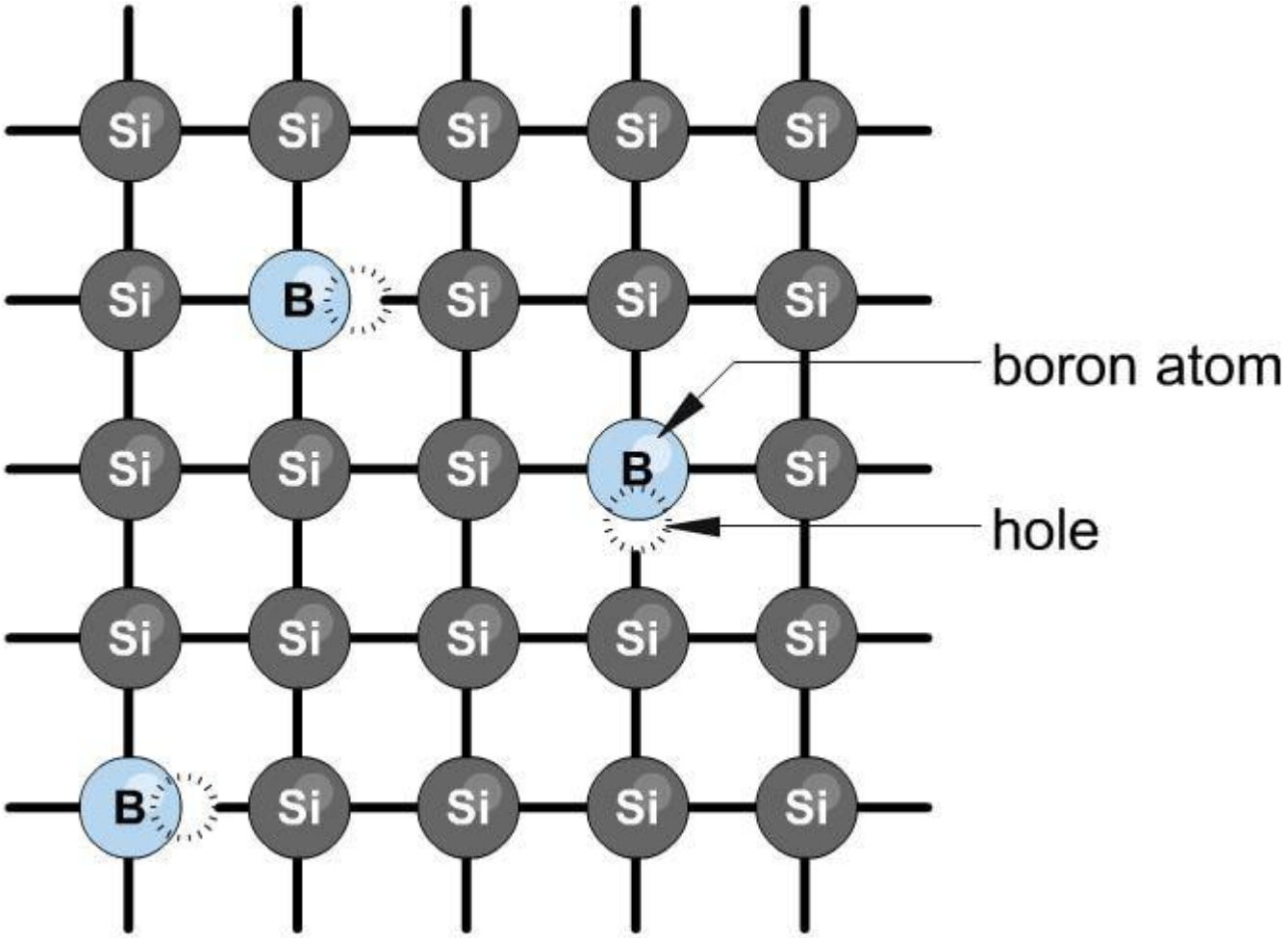
A) $1s^2\ 2s^2\ 2p^1$

B) $1s^2\ 2s^1\ 2p^2$

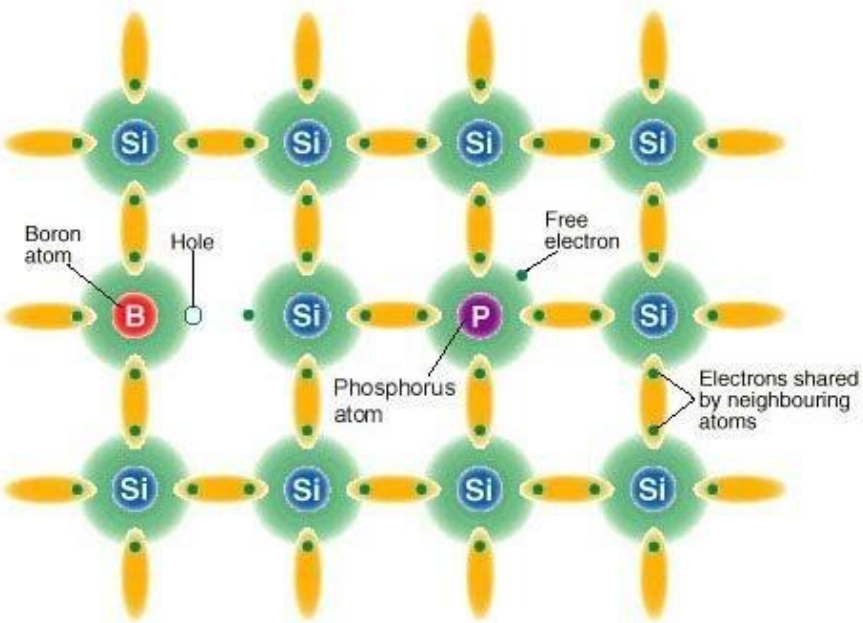
What will happen if 01 atom inside the Si crystal is replaced by 01 B atom? (*p-type Si in Si solar cell*)?

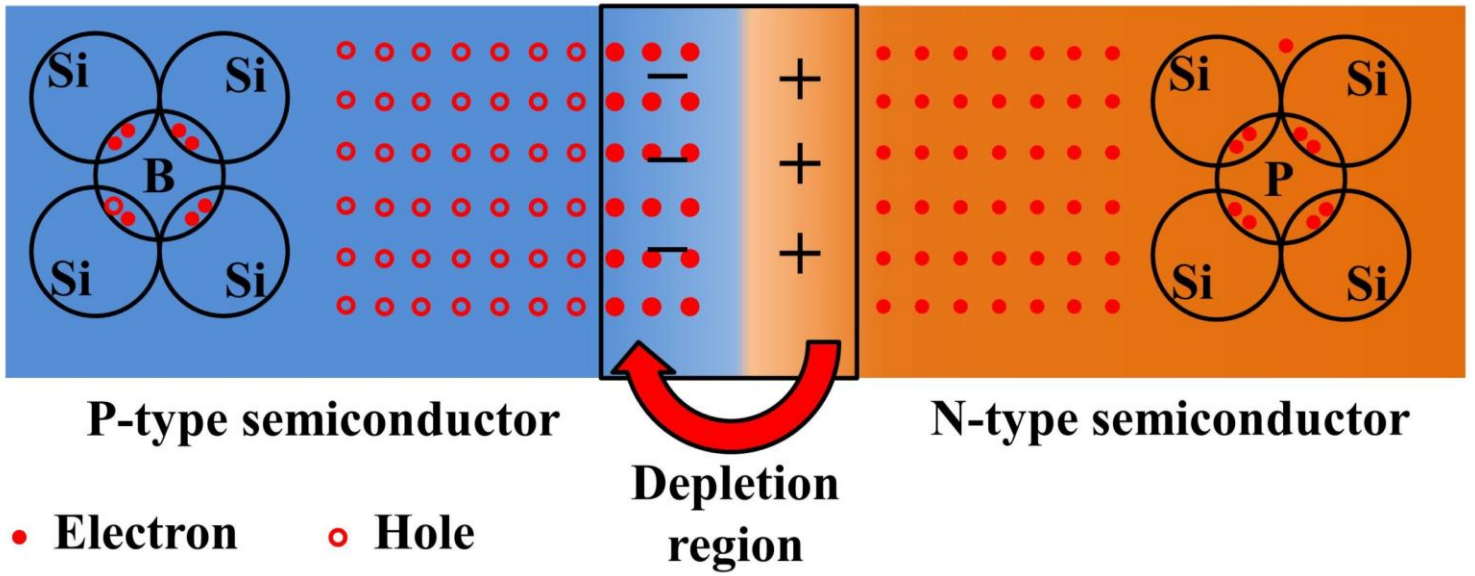


What will happen if 01 atom inside the Si crystal is replaced by 01 B atom? (*p-type Si in Si solar cell*)?

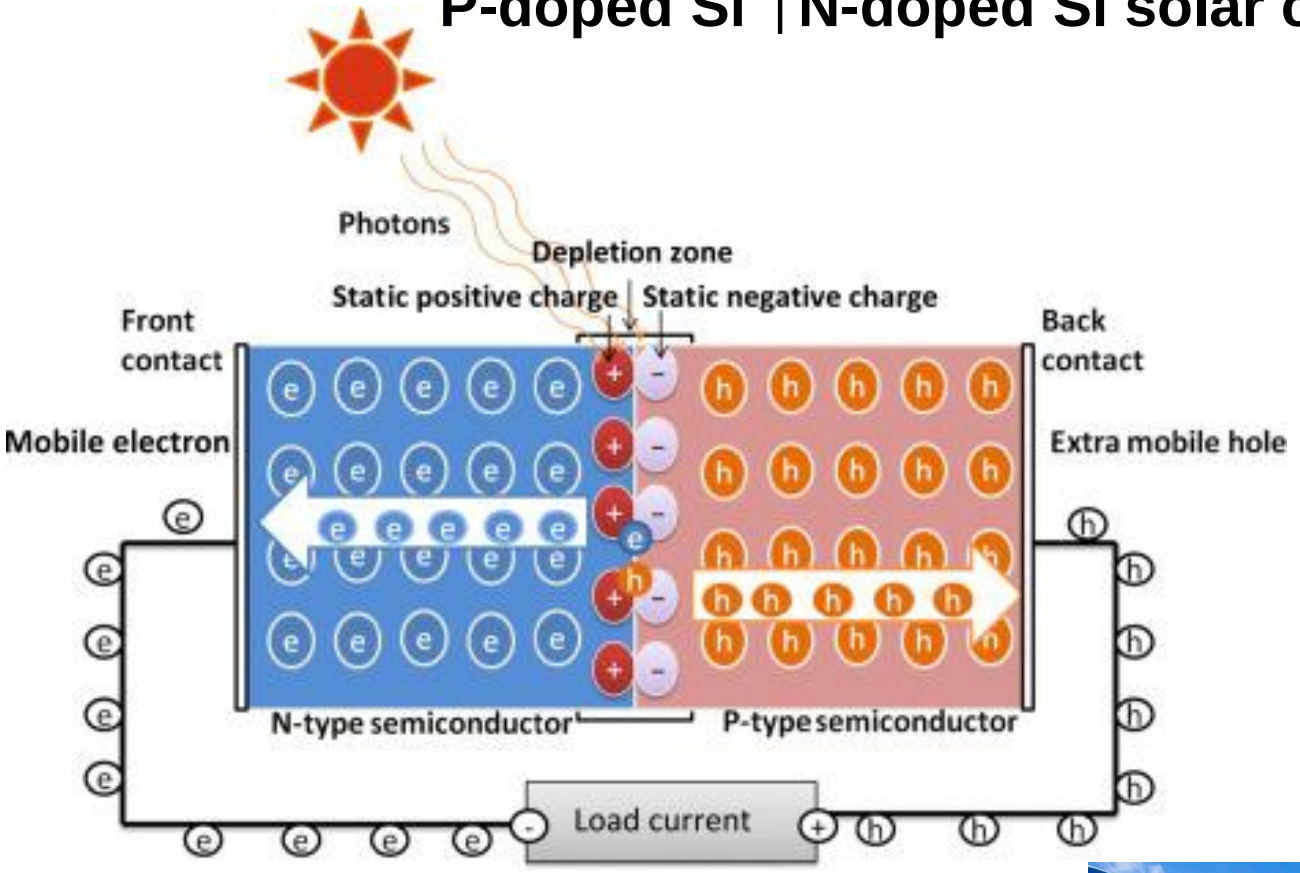


Replace 1 Si by 1 B >>> generate 1 extra hole
Replace 1 Si by 1 P >>> generate 1 extra electron



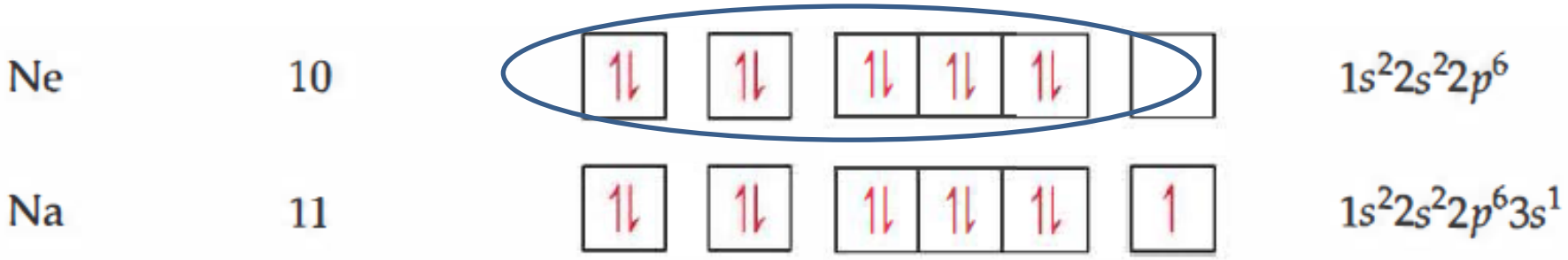


P-doped Si | N-doped Si solar cell





Condensed Writing



Use noble gas electron configuration to shorten other elements electron configuration

So we can write Na: $[Ne]3s^1$

8. THE PERIODIC TABLE

			Ti = 50	Zr = 90	? = 180	
			V = 51	Nb = 94	Ta = 182	
			Cr = 52	Mo = 96	W = 186	
			Mn = 55	Rh = 104,4	Pt = 197,4	
			Fe = 56	Ru = 104,4	Ir = 198	
			Ni = Co = 59	Pd = 106,6	Os = 199	
H = 1			Cu = 63,4	Ag = 108	Hg = 200	
	Be = 9,4	Mg = 24	Zn = 65,2	Cd = 112		
	B = 11	Al = 27,4	? = 68	Ur = 116	Au = 197?	
	C = 12	Si = 28	? = 70	Sn = 118		
	N = 14	P = 31	As = 75	Sb = 122	Bi = 210?	
	O = 16	S = 32	Se = 79,4	Te = 128?		
	F = 19	Cl = 35,5	Br = 80	J = 127		
Li = 7	Na = 23	K = 39	Rb = 85,4	Cs = 133	Tl = 204	
		Ca = 40	Sr = 87,6	Ba = 137	Pb = 207	
		? = 45	Ce = 92			
		?Er = 56	La = 94			
		?Yt = 60	Di = 95			
		?In = 75,6	Th = 118?			

Mendeleev 1869

based on « periodicity » of chemical properties

Elements having similar electronic configuration have similar properties

The cause of periodicity is the repetition of similar electronic configuration.

Elements with similar electronic structure of the outer shell (valence shell) : in the same column

Z=1	**H**	**Hydrogen**	**$1s^1$**
Z= 3	**Li**	**Lithium**	**$1s^2$ $2s^1$**
Z= 11	**Na**	**Sodium**	**$1s^2$ $2s^2$ $2p^6$ $3s^1$**
Z= 19	**K**	**Potassium**	**$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^1$**
Z= 37	**Rb**	**Rubidium**	**$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^{10}$ $4p^6$ $5s^1$**
Z = 55	**Cs**	**Cesium**	**[Xe]$6s^1$**

Elements with similar electronic structure of the outer shell (valence shell) : in the same column

Z= 4	**Be**	**Beryllium**	**$1s^2$ $2s^2$**
Z= 12	**Mg**	**Magnesium**	**$1s^2$ $2s^2$ $2p^6$ $3s^2$**
Z= 20	**Ca**	**Calcium**	**$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$**
Z= 38	**Sr**	**Strontium**	**$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^{10}$ $4p^6$ $5s^2$**

Elements with similar electronic structure of the outer shell (valence shell) : in the same column

Z= 2	**He**	**Helium**	**$1s^2$**
Z= 10	**Ne**	**Neon**	**$1s^2$ $2s^2$ $2p^6$**
Z= 18	**Ar**	**Argon**	**$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$**
Z= 36	**Kr**	**Krypton**	**$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^{10}$ $4p^6$**

Their outer shell is full (this is why He is here). « Noble gases »

Very stable electron configuration

Lines = progressive filling of subshells and shells (1s, 2s, 2p, 3s, 3p, 4s, 3d,…

2nd line :

Li	Be	B	C	N	O	F	Ne
$1s^2 2s^1$	$1s^2 2s^2$	$1s^2 2s^2 2p^1$	$1s^2 2s^2 2p^2$	$1s^2 2s^2 2p^3$	$1s^2 2s^2 2p^4$	$1s^2 2s^2 2p^5$	$1s^2 2s^2 2p^6$

These belong to « block s » =

Filling of the 2s orbital

2 electrons at most, 2 different elements

These belong to « block p » =

Filling of the 3 2p orbitals

6 electrons at most, 6 different elements

STRUCTURE OF THE PERIODIC CLASSIFICATION

d-block

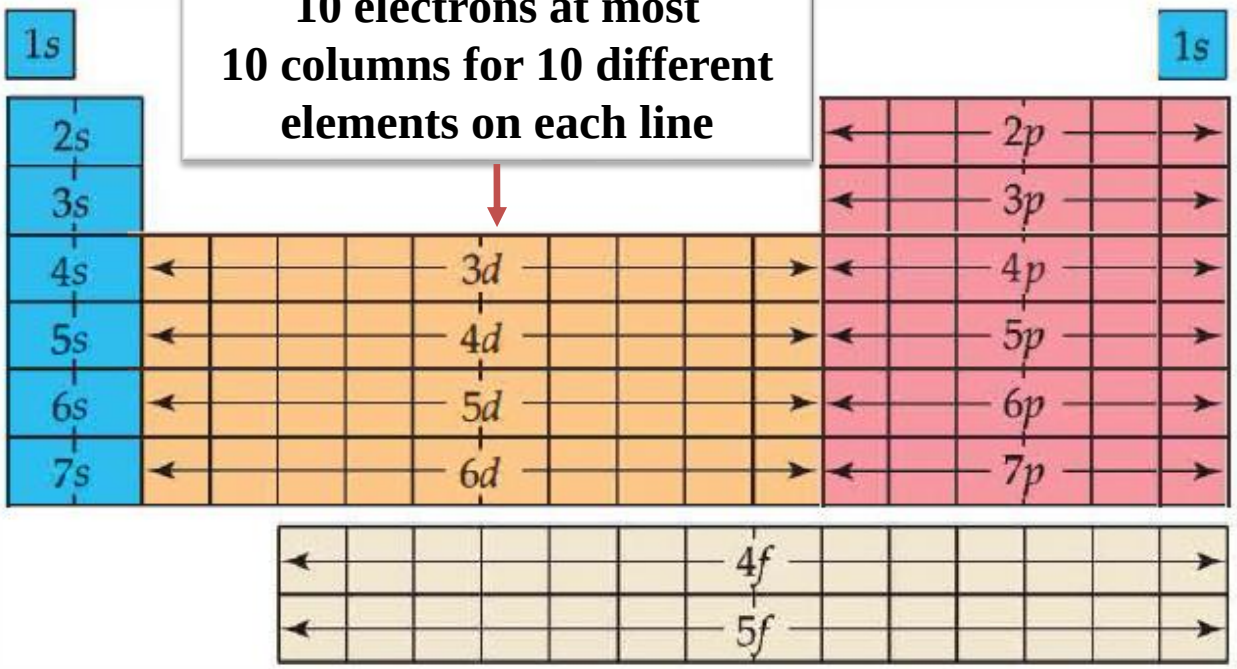
5 orbitals d
10 electrons at most
10 columns for 10 different elements on each line

He : exception

s-block

2 electrons at most

2 columns for 2 different elements on each line



p-block

3 orbitals p
6 electrons at most

6 columns for 6 different elements on each line

f-block **7 orbitals f, 14 electrons at most14 columns for 14 different elements on each line**

The electron configuration of $_{26}Fe$ is:

A) [Ar] $3d^6\ 4s^2$

B) [Ar] $4s^2\ 3d^6$

C) [Ar] $3d^8$

The electron configuration of $_{26}Fe^{2+}$ is:

A) [Ar] $3d^6$

B) [Ar] $4s^2\ 3d^4$

C) [Ar] $4s^1\ 3d^5$

The electron configuration of $_{26}Fe^{3+}$ is:

A) [Ar] $3d^5$

B) [Ar] $4s^2\ 3d^3$

The electron configuration of $_{27}Co$ is:

A) [Ar] $3d^7\ 4s^2$

B) [Ar] $4s^2\ 3d^7$

C) [Ar] $3d^9$

The electron configuration of $_{27}Co^{2+}$ is:

A) [Ar] $3d^7$

B) [Ar] $4s^2\ 3d^5$

C) [Ar] $4s^1\ 3d^6$

The electron configuration of $_{27}Co^{3+}$ is:

A) [Ar] $3d^6$

B) [Ar] $4s^1\ 3d^5$

General Chemistry I

Chemistry: The central science

Phong D. Tran

Department of Fundamental and Applied Sciences

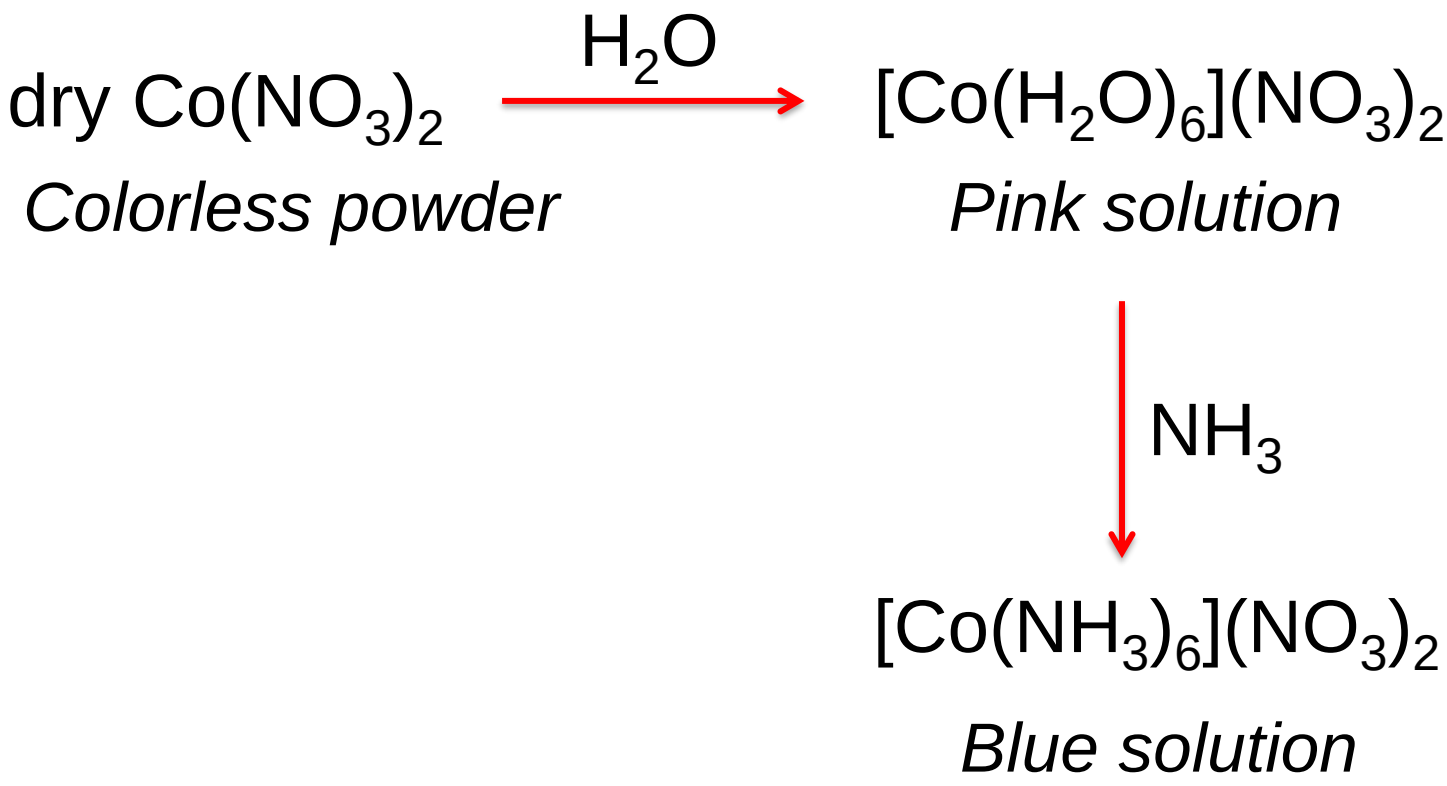
University of Science and Technology of Hanoi

Email: tran-dinh.phong@usth.edu.vn

Lectures 5,6: Electronic structure of atoms

Bonus…

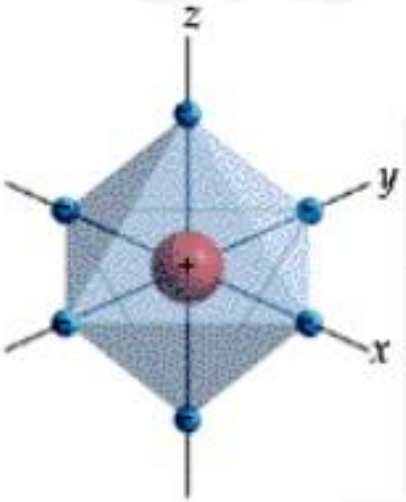
Experimental result

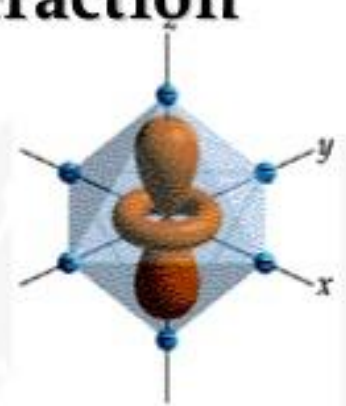


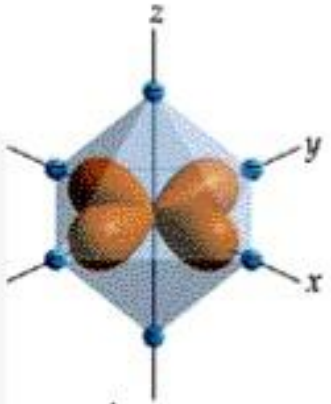
Can can we explain this phenomenon?

d-Orbitals and Ligand Interaction (Octahedral Field)

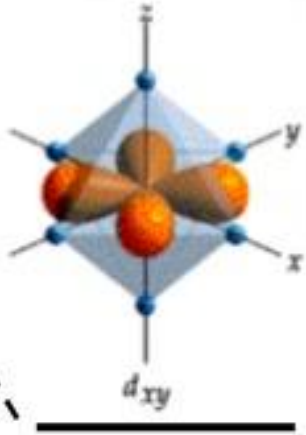
Ligands approach metal

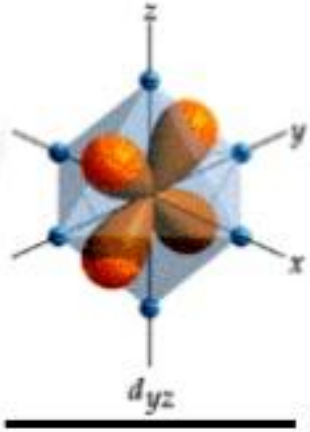


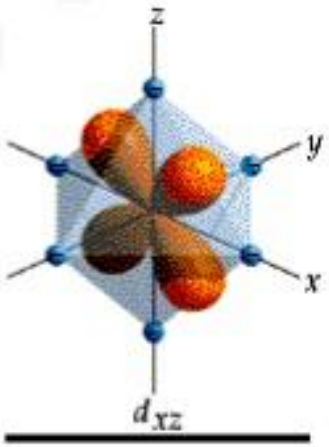




d-orbitals pointing directly at axis are affected most by electrostatic interaction

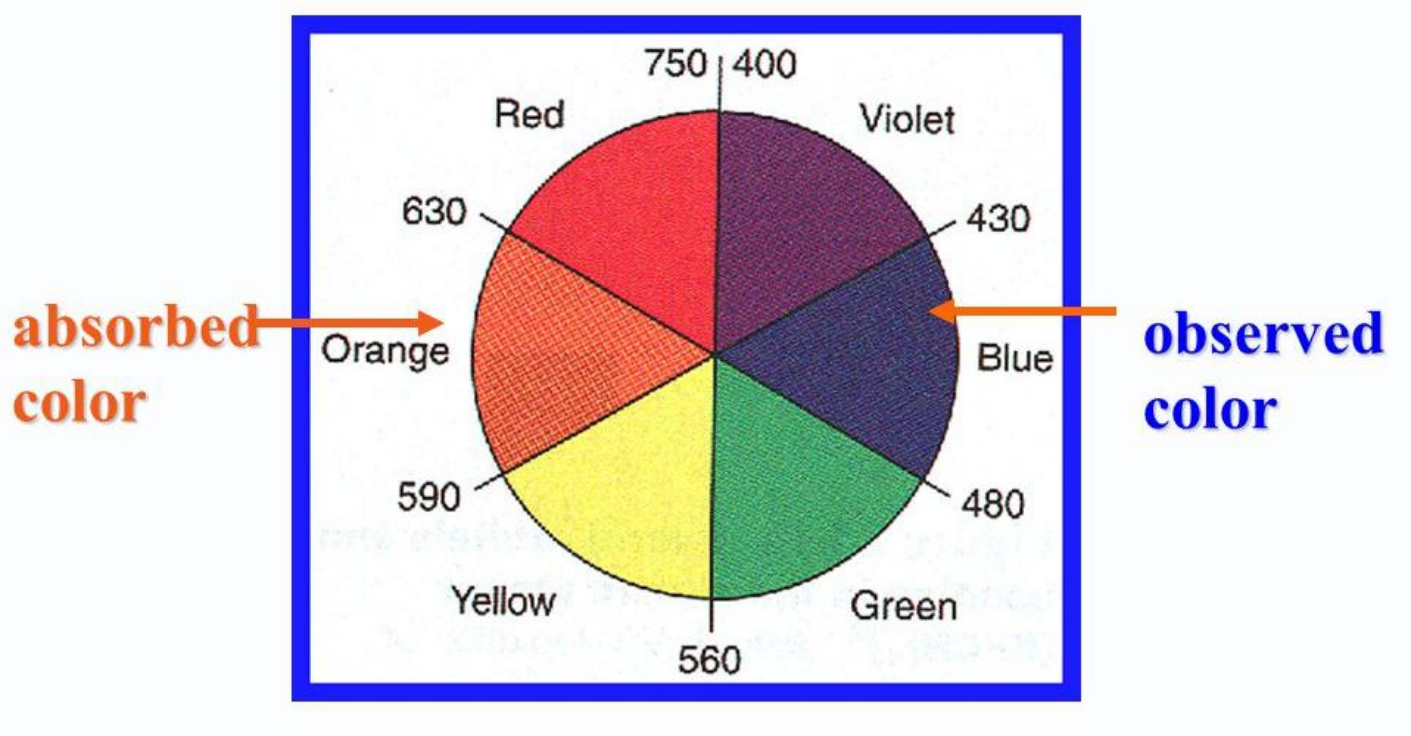






d-orbitals not pointing directly at axis are least affected (stabilized) by electrostatic interaction

Color: Absorption *versus* Observation



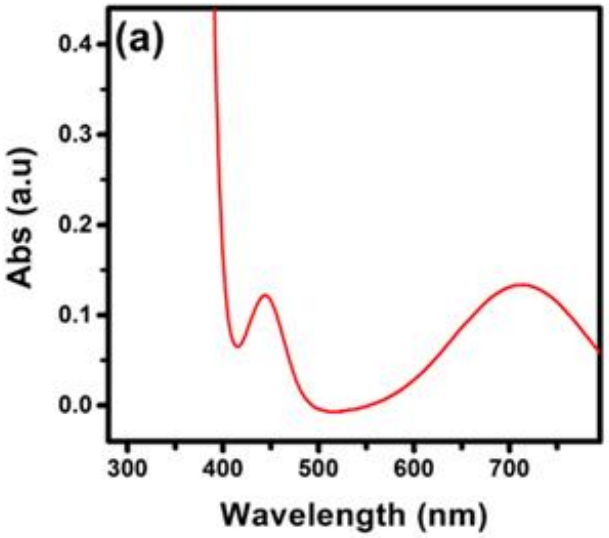
The colors of the visible light spectrum

Color	Wavelength interval	Frequency interval
Red	~ 700–635 nm	~ 430–480 THz
Orange	~ 635–590 nm	~ 480–510 THz
Yellow	~ 590–560 nm	~ 510–540 THz
Green	~ 560–520 nm	~ 540–580 THz

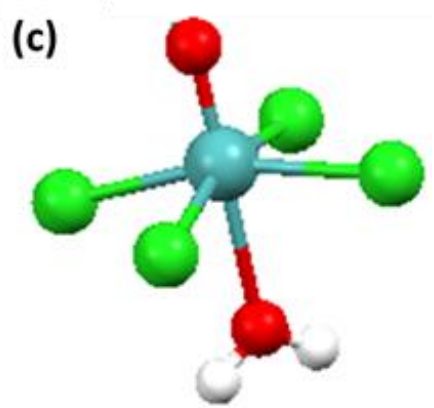
1. Write electronic configuration of following species:

Mo (Z=42), Mo^{5+} and Mo^{6+}

2. A compound of Mo shows green color. What could be the oxidation state of Mo within this compound? Mo^{5+} or Mo^{6+}? Give an explanation

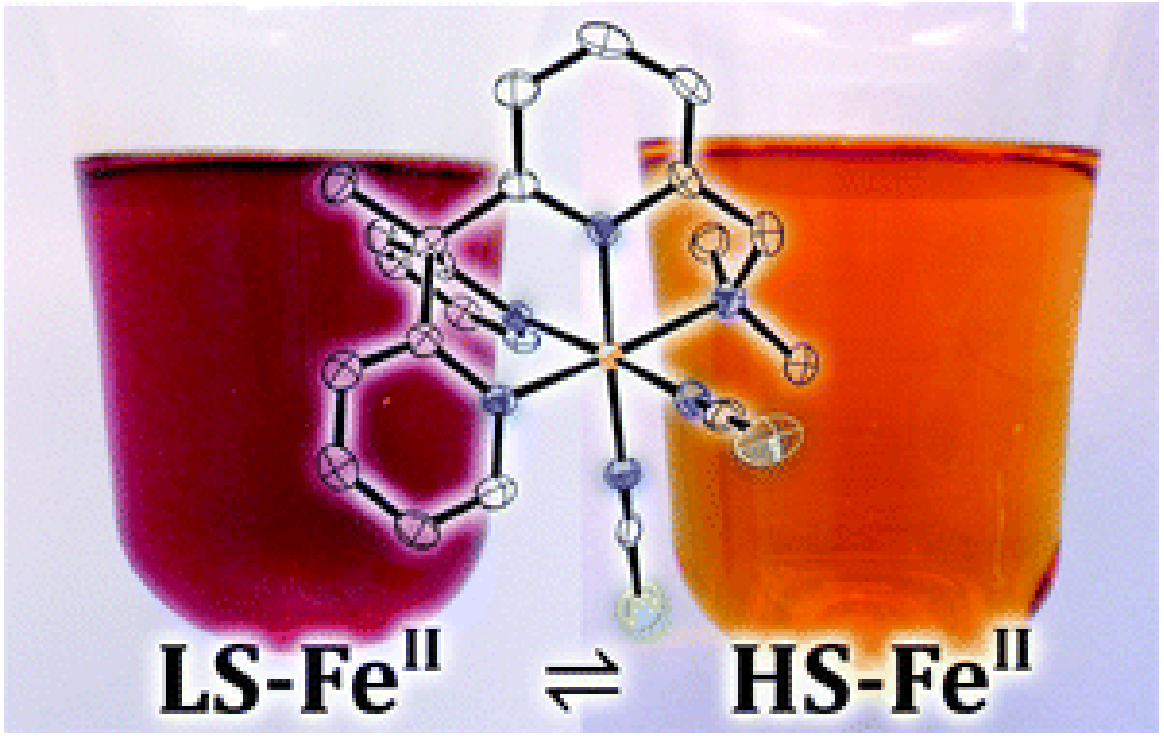




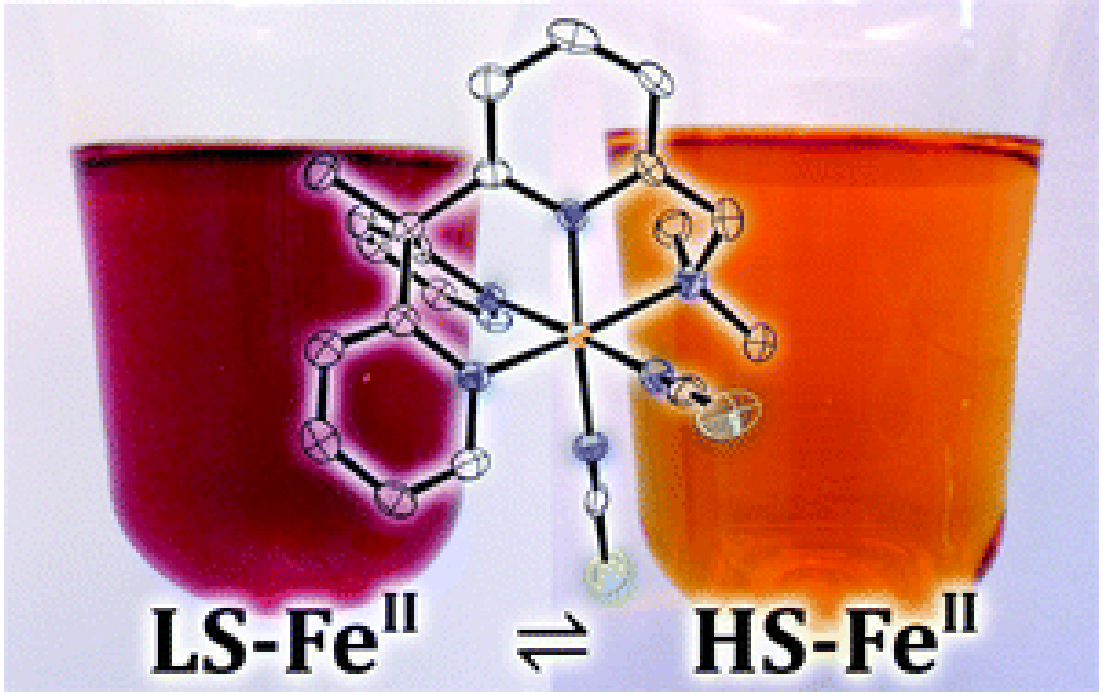


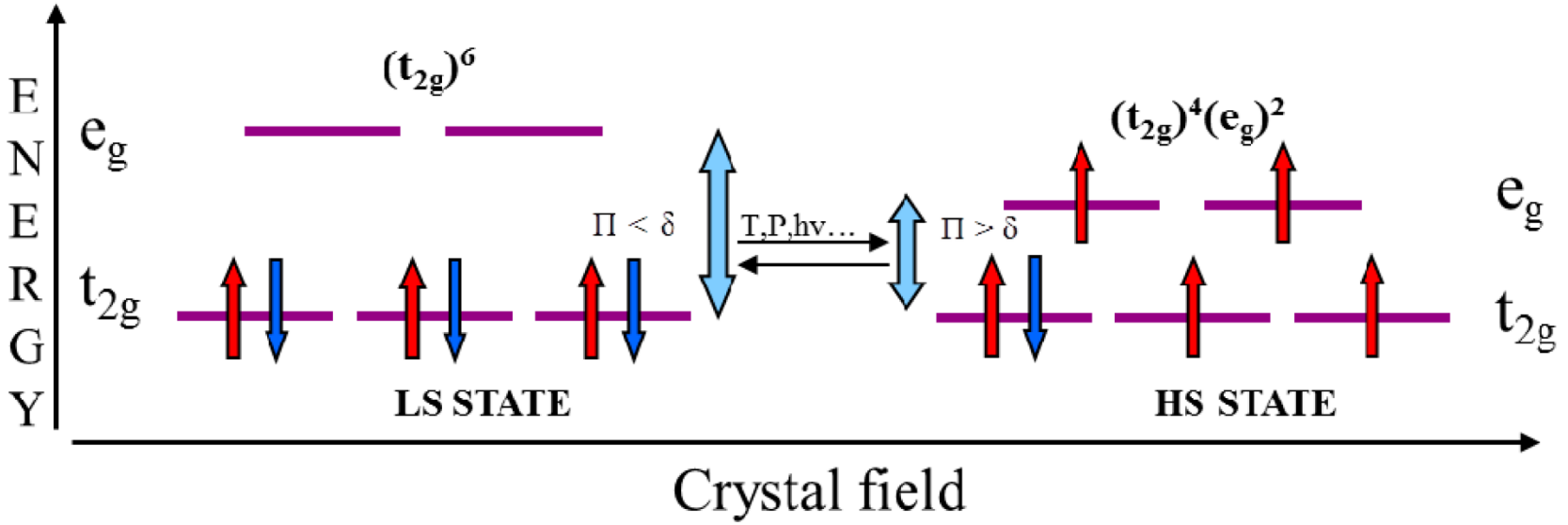
CECS@USTH data

1. Write down the electronic configuration of Fe (Z =26), Fe^{2+}, Fe^{3+}
2. A compound of Fe^{2+} (Fe^{II}) change color when it is heated up (*see figure*). Please give an explanation?

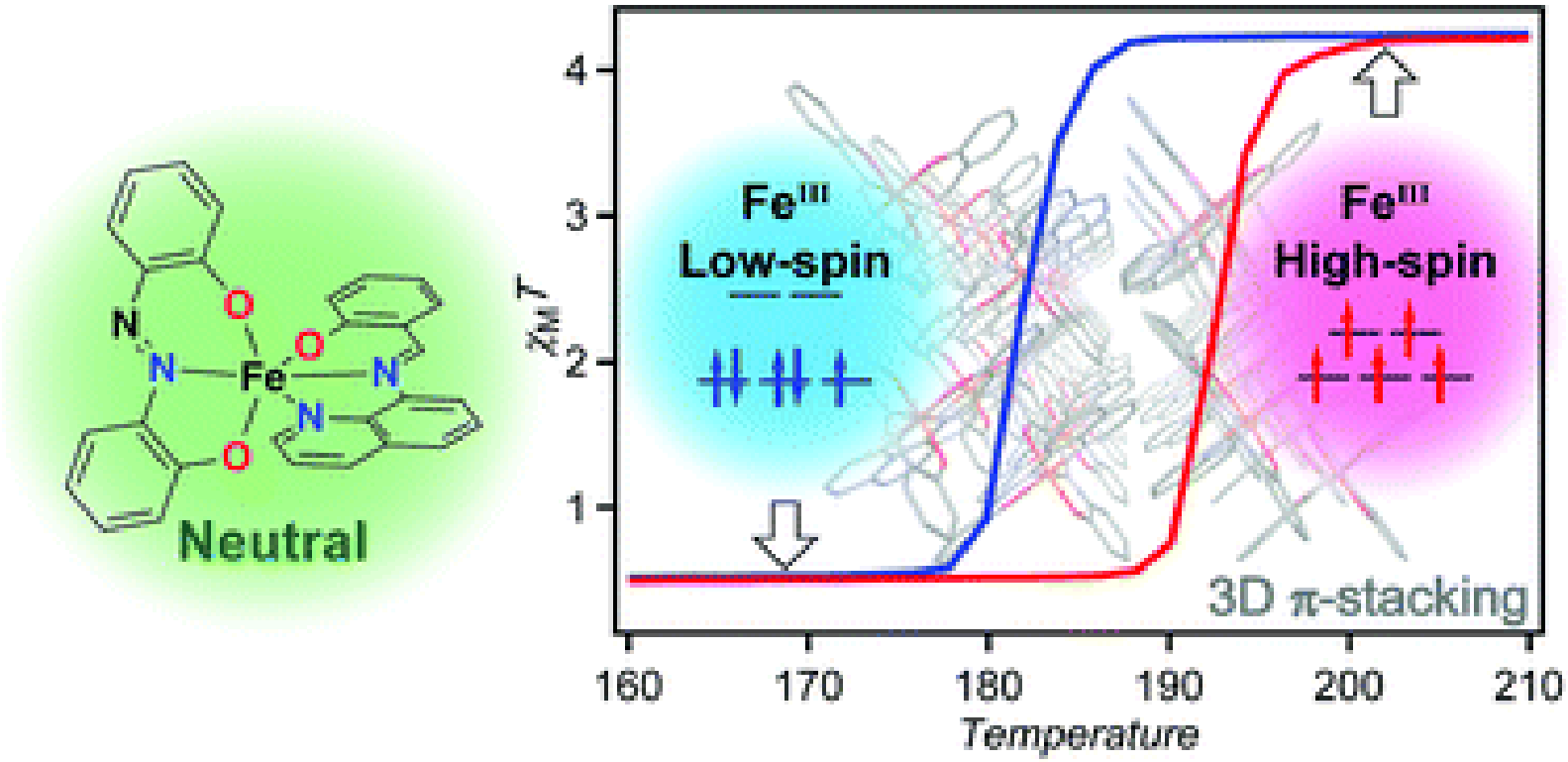


LS = Low Spin; HS = High Spin


LS-Fe^{II} ⇌ HS-Fe^{II}

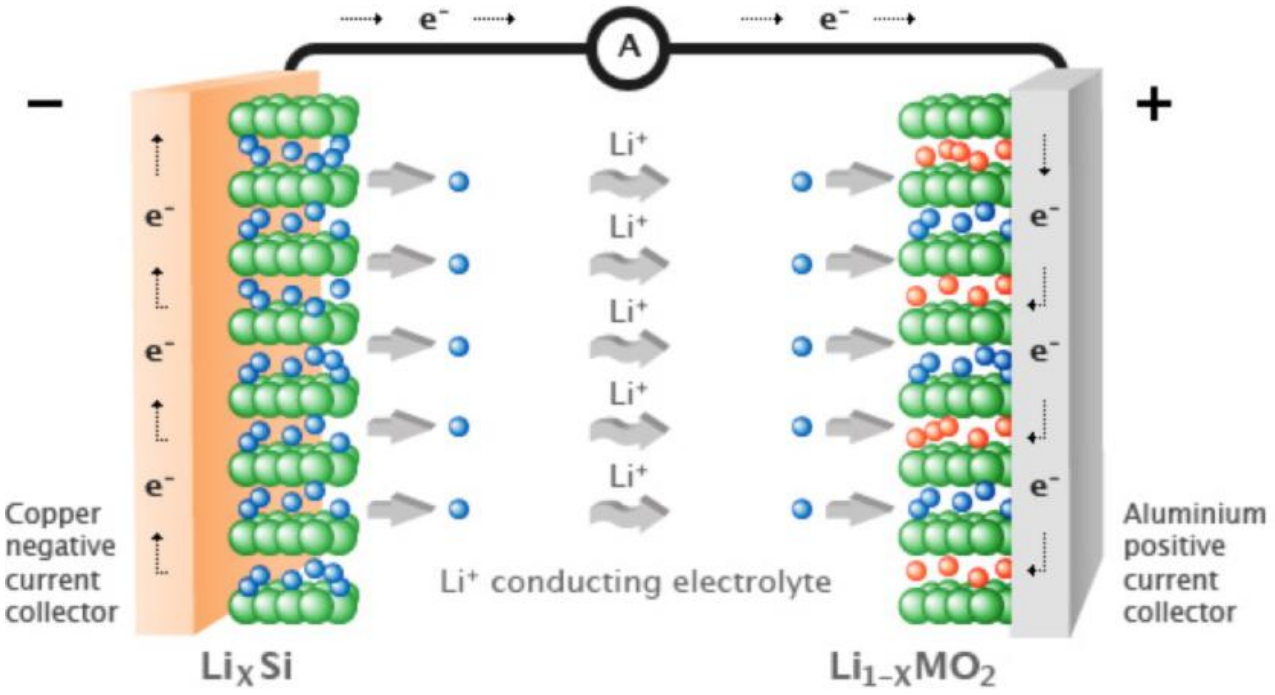

ENERGY
e_g
t_{2g}
$(t_{2g})^6$
Π < δ
T,P,hv…
Π > δ
$(t_{2g})^4(e_g)^2$
e_g
t_{2g}
LS STATE
HS STATE
Crystal field

Spin crossover: Information storage



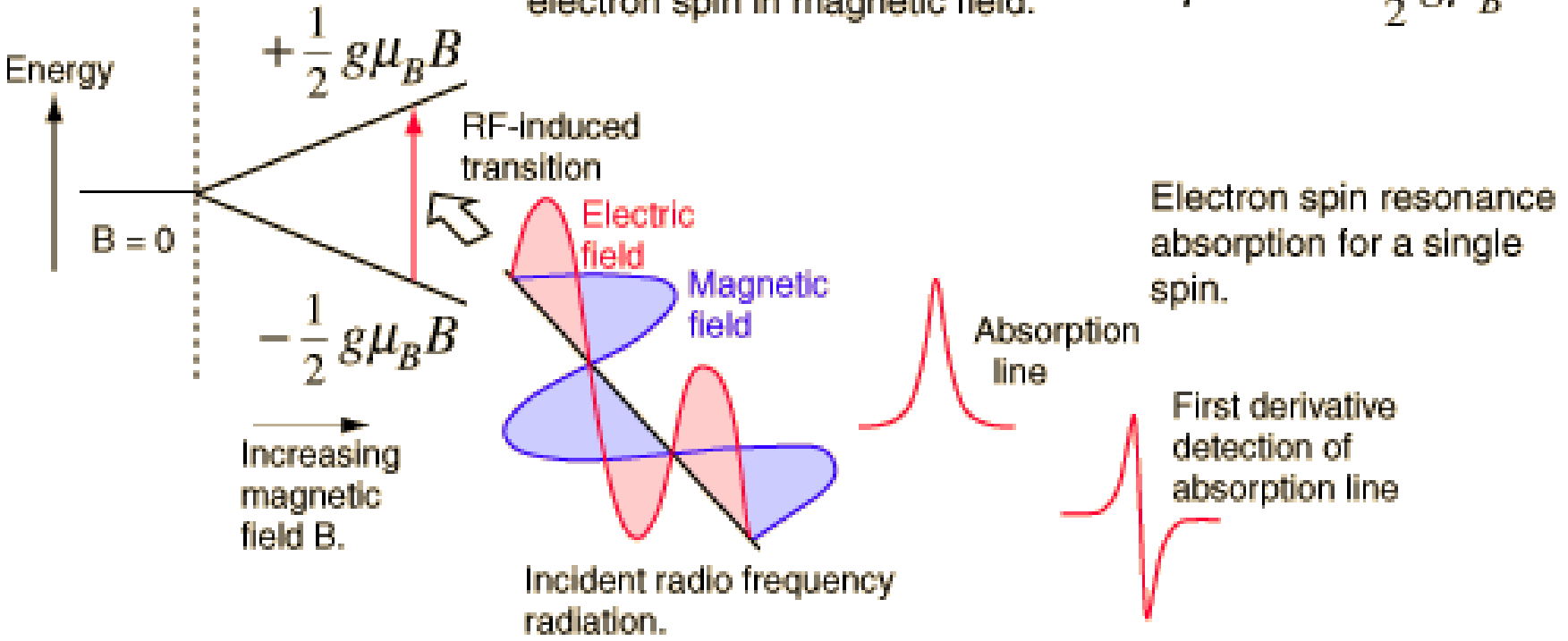
1. Write down electronic configuration of Li (Z=3), Na (Z=11), Mg (Z=12).
2. In which block these elements belong to?
3. Following is a schematic illustration of a Li battery design.

   Why is Li preferred for battery design?



Electron Paramagnetic Resonance: EPR

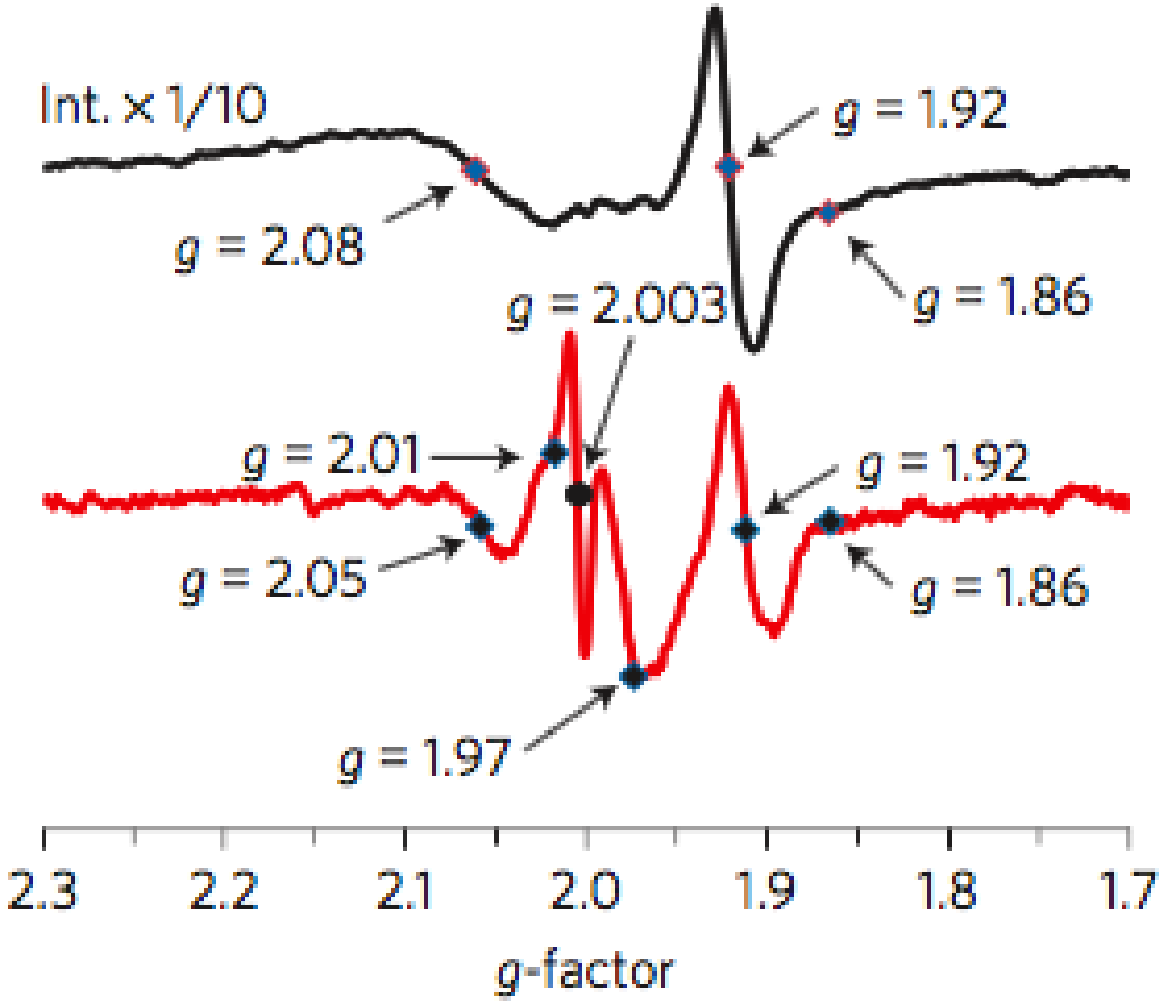
Magnetic potential energy of electron spin in magnetic field: $U = \mu \cdot B = \pm \frac{1}{2} g\mu_B B$





Only Mo^{V} species provide ERP signal.

Mo^{IV} and Mo^{VI} states are silent



5 VB 5B	6 VIB 6B	7 VIIB 7B
23 3407 V Vanadium 50.942	24 2671 Cr Chromium 51.996	25 2061 Mn Manganese 54.938
41 4744 Nb Niobium 92.906	42 4639 Mo Molybdenum 95.95	43 4265 Tc Technetium 98.907
73 5458 Ta Tantalum 180.948	74 5555 W Tungsten 183.85	75 5596 Re Rhenium 186.207